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Theoretical Foundations of Manifold Learning

Bachelor of Science Thesis by:

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Abstract

Geometry-based methods in machine learning have found increasing success in recent years for tasks such as dimensionality reduction and data visualization. The underlying assumption that most of these models share is that data points lie on a submanifold of low dimension embedded in a higher-dimensional space, or close to it.

In practice, it is not possible to observe the submanifold directly; thus, these methods approximate it using point clouds and the related adjacency graphs. The latter are used as a discrete structure that makes it possible to approximate the Laplace-Beltrami operator, a concept from Riemannian geometry that allows us to infer some geometrical properties of the manifold.

This thesis begins by reviewing some key notions from Riemannian geometry that are used in later sections. In particular, these concepts are used to prove the convergence of the graph Laplacian to the Laplace-Beltrami operator as the number of data points increases.

Later, the Laplacian Eigenmaps algorithm is introduced and discussed, showing how it can be used to find a low-dimensional representation of higher-dimensional data. Finally, this algorithm is applied to a dataset of rotating images, working under the assumption that the set of rotations forms a submanifold within the space of images.

Contents

1 Riemannian Manifold **2**

1.1 Laplace-Beltrami Operator 2

1.2 Heat Map 4

1.3 Geodesics 6

1.4 Volume Form 9

1.5 Covariant Derivatives, Curvature, Diameter 10

1.6 Spectrum of the Laplace-Beltrami Operator 14

2 Graph Laplacian and Manifold Learning **17**

2.1 Graph Laplacian 17

3 Point-wise convergence **18**

3.1 Proof of Point-wise Convergence 19

4 Spectral Convergence **25**

4.1 Proof of Spectral Convergence 26

5 Laplacian Eigenmaps **36**

5.1 The Algorithm 36

5.2 Justification 38

5.3 Results 43

6 Coil-100 Dataset **45**

6.1 Parameters and Choices for the Algorithm 46

6.2 Fortune Cat 47

6.3 Vase 48

6.4 Car Toy 49

A Appendix **50**

1 Riemannian Manifold

A Riemannian manifold $(\mathcal{M}, g)$ consists of a C^∞-manifold $\mathcal{M}$ and a Euclidean inner product g_p on each of the tangent spaces $T_p\mathcal{M}$ of $\mathcal{M}$. We also assume that $p \mapsto g_p$ varies smoothly on the points of $\mathcal{M}$. The metric g_p is usually referred to as Riemann metric.

Moreover all product spaces of the same dimension are isometric, thus all tangent spaces $T_p\mathcal{M}$ on a Riemannian manifold $(\mathcal{M}, g)$ are isometric to the n-dimensional Euclidean space $\mathbb{R}^n$ with its canonical inner product.

1.1 Laplace-Beltrami Operator

1.1.1 Standard Formulation

Let $\mathcal{M}$ be an n-dimensional, $n \geq 1$, connected, C^∞, Riemannian manifold. Given p on our manifold and a C^1 real-valued function f defined on a neighborhood of p, for each $\xi \in T_p\mathcal{M}$ we define the directional derivative of f at p in the direction ξ as ξf and we define it by:

$$\xi f = (f \circ \omega)'(0)$$

Where $\omega(t)$ is any path in M satisfying $\omega(0) = p$ and $\omega'(0) = \xi$. From this definition we can define the gradient of f to be the vector field on M such that:

$$\langle grad\, f, \xi \rangle = \xi\, f$$

In section 1.5.1 we will introduce the notion of covariant derivative, for now we only need to know that it is a rule that associates to each $p \in M$, $\xi \in T_pM$ and C^1 vector field X defined on a neighborhood of p, a vector $\nabla_\xi X \in T_pM$ such that:

$$\nabla_\xi(X + Y) = \nabla_\xi X + \nabla_\xi Y$$

$$\nabla_\xi(fX) = (\xi f)X(p) + f(p)\nabla_\xi X$$

where X, Y are C^1 vector fields, and f is a C^1 real valued function, all defined on a neighborhood of p. The vector $\nabla_\xi(X)$ is usually referred to as the covariant derivative of X with respect to ξ.

We can now define the divergence of a vector field X by letting ξ ranging over T_pM:

$$(div\, X)(p) = trace(\xi \mapsto \nabla_\xi X)$$

Finally, we can define the Laplacian of a function. For any C^k, $k \geq 2$ function f on M we define the function Laplacian of f (Δf) as:

$$\Delta f = div(grad\, f)$$

The Laplace-Beltrami operator, then generalizes on Riemannian manifolds the notion of Laplacian in the Euclidean space, which in general can be expressed as $\Delta f(\mathbf{x}) = \sum_i \frac{\partial^2 f}{\partial x_i^2}(\mathbf{x})$

1.1.2 Integral Formulation

Alternatively we could define the Laplace-Beltrami operator on a Riemannian Manifold distributionally. In this section we will denote ∇f to be the gradient of the function f on the manifold. This is the notation we will also use in the following sections.

Let $(\mathcal{M}, g)$ be a Riemannian Manifold. The Laplace-Beltrami operator is a second order differential operator acting on functions on $\mathcal{M}$.

The Laplacian is then defined using the Divergence Theorem and the integration by parts, and satisfies the following relation for any test function $\phi \in C_c^\infty(\mathcal{M})$ (ϕ is a smooth function with compact support):

$$\int_{\mathcal{M}} (\Delta f)\, \phi \; d\mu = -\int_{\mathcal{M}} \langle \nabla f, \nabla \phi \rangle_g \; d\mu$$

Where $f \in C^\infty(\mathcal{M})$ is a smooth function on $\mathcal{M}$, ∇ is the gradient operator (with respect to the metric g), and $d\mu$ is the volume form, which we will define in the following sections.

Defining the Laplace-Beltrami operator via integration by parts, we have the same result as before:

$$\Delta f = div(\nabla f)$$

It is to keep in mind, however, that in many applications the Laplace-Beltrami operator is defined with opposite sign:

$$\Delta f = -div(\nabla f)$$

This is usually needed in order to have that the eigenvalues of Δ are all non-negative.

1.2 Heat Map

The Laplace Beltrami operator is closely related to the heat flow, so we want to study the heat equation and its solutions. We say that a sufficiently differentiable function $u(\mathbf{x}, t)$ satisfies the heat equation if

$$\frac{\partial}{\partial t}u(\mathbf{x}, t) - \Delta u(\mathbf{x}, t) = 0$$

The heat equation describes diffusion of heat with the initial distribution $u(\mathbf{x}, t)$.

Notice that, as we define the Laplacian with opposite sign to ensure positive spectrum ($\Delta f = -div(\nabla f)$), then the heat equation will be

$$\frac{\partial}{\partial t}u + \Delta u = 0$$

The solution to the heat equation is given by a semigroup of heat operators $\mathbf{H}^t$. Given an initial heat distribution $f(\mathbf{x})$, $\mathbf{H}^t(f)$ is the heat distribution (heat operator) at time t.

It turns out that this operator is given by convolution with the heat kernel, which for $\mathbb{R}^k$ is the usual Gaussian:

$$\mathbf{H}^t f(\mathbf{x}) = \int_{\mathbb{R}^k} f(\mathbf{y}) H^t(\mathbf{x}, \mathbf{y}) d\mathbf{y}$$

$$H^t(\mathbf{x}, \mathbf{y}) = (4\pi t)^{-\frac{k}{2}} e^{-\frac{||\mathbf{x}-\mathbf{y}||^2}{4t}}$$

In particular, notice that the heat kernel $H^t(\mathbf{x}, \mathbf{y})$ can be always defined on any compact manifold without boundary.

Theorem. Let $f(\mathbf{x})$ be a sufficiently differentiable bounded function. We then have:

$$\mathbf{H}^t f(x) = (4\pi t)^{-\frac{k}{2}} \int_{\mathbb{R}^k} e^{-\frac{||\mathbf{x}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mathbf{y}$$

$$f(\mathbf{x}) = \lim_{t\to 0} \mathbf{H}^t f(\mathbf{x}) = \lim_{t\to 0} (4\pi t)^{-\frac{k}{2}} \int_{\mathbb{R}^k} e^{-\frac{||\mathbf{x}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mathbf{y}$$

The function $u(\mathbf{x}, t) = \mathbf{H}^t f$ satisfied the heat equation: $\frac{\partial}{\partial t}u(\mathbf{x}, t) + \Delta u(\mathbf{x}, t) = 0$.

Thus the heat equation can be used to approximate the Laplace operator. Now let's notice that:

$$-\Delta f(\mathbf{x}) = \frac{\partial}{\partial t}\mathbf{H}^t f(\mathbf{x})\bigg|_{t=0} =$$

$$\lim_{t\to 0} \frac{1}{t}\left((4\pi t)^{-k/2} \int_{\mathbb{R}^k} e^{-\frac{||\mathbf{x}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mathbf{y} - f(\mathbf{x})(4\pi t)^{-k/2} \int_{\mathbb{R}^k} e^{-\frac{||\mathbf{x}-\mathbf{y}||^2}{4t}} d\mathbf{y} \right)$$

As this quantity can be approximated by the point cloud by using summations instead of integrals, we can approximate the Laplacian as:

$$\hat{\Delta} f(\mathbf{x}) = \frac{1}{t}\frac{(4\pi t)^{-k/2}}{n}\left(f(\mathbf{x}) \sum_i e^{-\frac{||\mathbf{x_i}-\mathbf{x}||^2}{4t}} - \sum_i e^{-\frac{||\mathbf{x_i}-\mathbf{x}||^2}{4t}} f(\mathbf{x_i}) \right)$$

$$= \frac{(4\pi t)^{-\frac{k+2}{2}}}{n} \mathbf{L}_n^t(f)(\mathbf{x})$$

Where we have defined $\hat{\mathbf{L}}_n^t(f)(\mathbf{x}) = \frac{1}{t}\left(f(\mathbf{x}) \sum_i e^{-\frac{||\mathbf{x_i}-\mathbf{x}||^2}{4t}} - \sum_i e^{-\frac{||\mathbf{x_i}-\mathbf{x}||^2}{4t}} f(\mathbf{x_i}) \right)$

In the later sections we will use different definitions for this same operator; sometimes, for simplicity, it's easier to define it with or without the scaling constant $\frac{(4\pi t)^{-k/2}}{n}$. In each section it will be clear how we define it in that context, but in general it always serves as an approximation for the Laplace-Beltrami operator.

1.2.1 Exponential Form of Solution

Given the heat equation $\frac{\partial u}{\partial t} = \Delta u$ and initial distribution $u(x,0) = f(x)$, we want to solve the initial value problem:

$$\begin{cases} \frac{\partial u}{\partial t} = \Delta u \\ u(x,0) = f(x) \end{cases}$$

This will tell us that a solution of the heat equation can be written as

$$\mathbf{H}^t(f) = e^{t\Delta} f$$

Here $e^{t\Delta}$ is the bounded operator on $L^2(\mathcal{M})$ obtained via spectral calculus, where it is defined through the functional calculus for self-adjoint operators.
The proof of this statement can be found in [7] using basic theory of PDEs. Notice that, whenever we define the Laplace-Beltrami operator as $\Delta f = -div(\nabla f)$, we get that the solution is $e^{-t\Delta}$ and we will write $\mathbf{H}^t = e^{-t\Delta}$.

1.3 Geodesics

A geodesic on a manifold is a curve that represents the shortest path between two points on the manifold, locally. In fact, geodesics are the equivalent for manifolds of straight lines in the Euclidean space.

Defining the acceleration of a curve $c : I \to \mathcal{M}$ by the formula

$$c'' = \frac{d^2c}{dt^2}$$

we can define a geodesic curve, more formally, as follows.

A geodesic is a C^∞ curve $c : I \to \mathcal{M}$ with vanishing acceleration $c'' = 0$. If c is a geodesic, then $|c'| = \sqrt{g(c', c')}$ is constant. In particular if c is parametrized by arclength, then $|c'| = 1$.

Assume we have a compact differentiable k-dimensional submanifold of $\mathbb{R}^N$ with the induced Riemann structure. That means that we have a notion of length for curves on $\mathcal{M}$. Given two points $\mathbf{x}, \mathbf{y}$ the geodesic distance

$$dist_{\mathcal{M}}(\mathbf{x}, \mathbf{y}) \geq ||\mathbf{x} - \mathbf{y}||$$

Moreover the difference between geodesic distance ($dist_{\mathcal{M}}(\mathbf{x}, \mathbf{y})$) and chordal distance ($||\mathbf{x} - \mathbf{y}||$) is very small when $\mathbf{x}$ and $\mathbf{y}$ are very close. In particular:

$$dist_{\mathcal{M}}(\mathbf{x}, \mathbf{y}) = ||\mathbf{x} - \mathbf{y}|| + O(||\mathbf{x} - \mathbf{y}||^3)$$

Given a point $\mathbf{p} \in \mathcal{M}$ one can identify the tangent space $T_{\mathbf{p}}\mathcal{M}$ with an affine subspace of $\mathbb{R}^N$ passing through $\mathbf{p}$. This space has a natural linear structure with origin at $\mathbf{p}$.

It is possible to define the exponential map

$$exp_{\mathbf{p}} : T_{\mathbf{p}}\mathcal{M} \to \mathcal{M}$$

as

$$\exp_{\mathbf{p}}(v) = \gamma_v(1),$$

where $\gamma_v(t)$ is the unique geodesic starting at $\mathbf{p}$ with initial velocity v, i.e.,

$$\gamma_v(0) = \mathbf{p}, \quad \dot{\gamma}_v(0) = v.$$

Thus, the exponential map sends a vector $v \in T_{\mathbf{p}}\mathcal{M}$ to the point reached at time $t = 1$ along the geodesic with initial velocity v.

Hence the exponential map takes lines through origin in $T_{\mathbf{p}}\mathcal{M}$ to geodesics passing through $\mathbf{p}$. The exponential map is a local diffeomorphism and produces a natural system of coordinates for some neighborhood of $\mathbf{p}$.

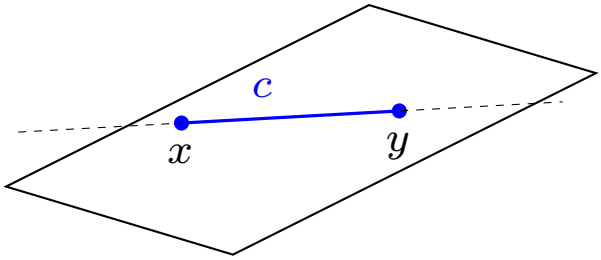
1.3.1 Examples

a) Geodesics in the Euclidean Space

As already mentioned, geodesics in the Euclidean space are simply straight lines through a point. In fact, the shortest path between two points in $\mathbb{R}^n$ is a straight line.
Thus, the parametrization of c would be:

$$c(t) = p + t \cdot v$$

Where p is a point on the line and v is the direction of the curve.

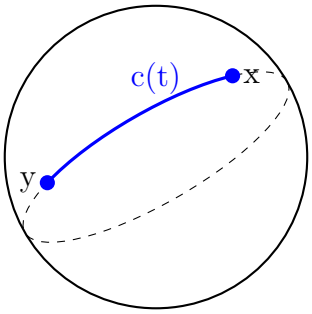


b) Geodesics on a Sphere

Let us now consider the two dimensional sphere S^2. The shortest path between two points on the sphere lies along a great circle (a circle on the surface of a sphere that has the same radius as the sphere itself, and its center coincides with the center of the sphere).
To develop intuition about great circles, consider the sphere S^2 to be the earth. Then the equator and any line of longitude are great circles. In particular the parametrization of the great circle will be given by the intersection of the sphere ($x^2 + y^2 + z^2 = R^2$) and a plane through the origin ($ax + by + cx = 0$). If we have a basis of two $\mathbf{u}, \mathbf{v}$ belonging to the plane, then the parametrization of the great circle will be:

$$c(t) = R\,cos(t)\mathbf{u} + R\,sin(t)\mathbf{v}$$

with $t \in [0, 2\pi)$



1.4 Volume Form

As in the Euclidean space the inner product allows to define norms, angles, areas and volumes, the same thing can be done on Riemannian manifolds using the Riemannian metric g.

In an oriented Riemannian n-dimensional manifold $(\mathcal{M}, g)$ the volume form is defined to be an n-form on $\mathcal{M}$. In particular to compute the volume of a parallelepiped spanned by n vectors $v_1, v_2, ..., v_n \in T_p\mathcal{M}$ (tangent vectors at point $p \in \mathcal{M}$), the volume form will be:

$$vol_g(v_1, ..., v_n) = det[g(v_i, e_j)]$$

where $e_1, ..., e_n$ is any positively oriented orthonormal basis of $T_p\mathcal{M}$ and $g(v_i, e_j)$ denotes the entries of the $n \times n$ matrix with entry ij the Riemannian metric evaluated on the vectors v_i, e_j.

The volume form will induce a measure $d\mu$ (or $d\mu_g$) that will allow us to define a weighted volume of a function f over a manifold. Denoting g_{ij} the matrix $g_{ij} = g(\frac{d}{dx_i}, \frac{d}{dx_j})$ we will have that the induced measure is defined as:

$$d\mu_g = \sqrt{det(g_{ij})}\, dx_1 \dots dx_n$$

the weighted volume of a function f over a manifold will be computed as:

$$\int_{\mathcal{M}} f \, d\mu_g$$

.

The induced measure can also be used to compute the volume of a subset of the manifold $A \in \mathcal{M}$ (and of course, also of the whole manifold) as:

$$vol(A) = \int_A d\mu_g$$

Unlike the volume form, the volume measure always exists, even when the manifold is not orientable.

1.5 Covariant Derivatives, Curvature, Diameter

1.5.1 Covariant Derivatives

As already mentioned in the standard definition of Laplace-Beltrami operator, covariant derivative provides a way to differentiate vector fields on a Riemannian manifold while respecting its geometric structure.

Let $(\mathcal{M}, g)$ be a Riemannian manifold. A connection on $\mathcal{M}$ is a rule that assigns to each pair of vector fields X, Y on the manifold, another vector field $\nabla_X Y$, called the covariant derivative of Y along X. A connection ∇ is said to be a Riemannian connection if it satisfies the following two properties:

- Metric Compatibility: for any vector fields X, Y, Z, we have

$$X \cdot g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

  This ensures that the inner product is preserved under parallel transport.

- Torsion-Free Property: The connection satisfies

$$\nabla_X Y - \nabla_Y X = [X, Y],$$

  meaning that the torsion tensor $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$ vanishes.

These two properties uniquely define the Levi-Civita connection, which is the natural connection associated with a Riemannian metric.

1.5.2 Curvature

A useful concept in the study of manifolds is the one of curvature, which describes how a Riemannian manifold deviates from being flat (i.e. Euclidean). Hence it gives an idea of how the shape changes locally. We will first introduce the curvature tensor and then some different types of curvature; different notions of curvature provide different levels of geometric information. In particular the sectional curvature will be useful in the later sections to prove the point wise convergence to Δ of its functional approximation.

a) Curvature Tensor

The Riemann curvature tensor measures the noncommutativity of covariant derivatives and is defined as a (1,3)-tensor:

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

Equivalently, it can be written using Lie brackets as:

$$R(X,Y)Z = [\nabla_X, \nabla_Y]Z - \nabla_{[X,Y]}Z.$$

For a given vector $v \in T_p\mathcal{M}$, we define the directional curvature operator as:

$$R_v(w) = R(w,v)v : T_p\mathcal{M} \to T_p\mathcal{M}.$$

This operator will be useful in defining sectional curvature.

b) Sectional Curvature

The sectional curvature gives a measure of the curvature associated with a two-dimensional plane in the tangent space. Given a plane spanned by two linearly independent vectors $v, w \in T_p\mathcal{M}$, the sectional curvature is defined as:

$$sec(v,w) = \frac{g(R_v(w),w)}{g(v,v)g(w,w) - g(v,w)^2}.$$

This quantity only depends on the plane spanned by v and w, not on the choice of basis (in particular the denominator corresponds to the squared area of the parallelogram spanned by v and w).

Example:

- In $\mathbb{R}^n$ with the Euclidean metric, the sectional curvature is identically zero.
- In the standard n-sphere $\mathbb{S}^n$ with the induced metric, the sectional curvature is constant and equal to 1.

c) Ricci Curvature

The Ricci curvature is obtained by tracing the curvature tensor. Given an orthonormal basis $\{e_1, ..., e_n\}$ of $T_p\mathcal{M}$, the Ricci tensor is defined as:

$$Ric(v,w) = tr(x \mapsto R(x,v)w)$$

$$= \sum_{i=1}^{n} g(R(e_i,v)w, e_i).$$

Alternatively, it can be written as a $(1,1)$-tensor:

$$Ric(v) = \sum_{i=1}^{n} R(v,e_i)e_i.$$

Example:

- In a space of constant sectional curvature sec, the Ricci tensor satisfies $Ric = (n-1)\, sec\ g$.
- A sphere of radius r has Ricci Curvature, $Ric = \frac{(n-1)}{r^2}$

d) Scalar Curvature

The scalar curvature is a global measure of curvature obtained by taking the trace of the Ricci tensor:

$$scal = \sum_{i=1}^{n} g(Ric(e_i), e_i).$$

This function assigns a real number to each point of the manifold and provides information about the average curvature at that point.

Example:

- In the standard n-sphere S^n, the scalar curvature is $scal = n(n-1)$.

1.5.3 Diameter of a Manifold

The diameter of a connected Riemannian manifold $(\mathcal{M}, g)$ is a measure of its largest possible distance. It is defined as:

$$diam(\mathcal{M}) = \sup_{x,y \in \mathcal{M}} d(x, y),$$

where $d(x, y)$ is the Riemannian distance between x and y, given by the infimum of the length of all piecewise smooth curves connecting them:

$$d(x, y) = \inf_{\gamma} \int_0^1 \|\dot{\gamma}(t)\| dt.$$

Notice that for a complete manifold (i.e. all geodesics are defined for all time), this distance is achieved when γ is a geodesic curve. As we will deal mainly with compact manifolds (which are always complete), the distance between x and y on the manifold can simply be defined as the length of the geodesic connecting x and y.

Example:

- The diameter of the sphere of radius r $\mathbb{S}^n_r$ is $\pi \cdot r$.
- The diameter of an n-dimensional Euclidean space $\mathbb{R}^n$ is infinite.

1.5.4 Bonnet-Myers Theorem

We now state an important link between Ricci Curvature and Diameter of the manifold through the Bonnet-Myers theorem.

Theorem Let $(\mathcal{M}, g)$ be a complete and connected Riemannian manifold of dimension n whose Ricci curvature satisfies the inequality

$$\mathrm{Ric}_p(v) \geq (n-1)K$$

for every $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$ of unit length. Then, the manifold $\mathcal{M}$ is compact, and its diameter satisfies the bound

$$\mathrm{diam}(\mathcal{M}) \leq \frac{\pi}{\sqrt{K}}.$$

That is, any two points in $\mathcal{M}$ can be joined by a geodesic segment of length at most πr.

Moreover, the bound on the diameter is sharp, and equality is obtained for the sphere of radius $\frac{1}{\sqrt{K}}$. In fact, such sphere has constant Ricci curvature equal to $(n-1)K$ and diameter $\pi/\sqrt{K}$.

1.6 Spectrum of the Laplace-Beltrami Operator

The spectrum of the Laplace-Beltrami Δ operator on a Riemannian Manifold $(\mathcal{M}, g)$ consists of eigenvalues $\lambda \in \mathbb{R}$ and eigenfunctions $f : \mathcal{M} \to \mathbb{R}$ that satisfy the equation:

$$\Delta f = \lambda f$$

Eigenfunctions of the Laplace-Beltrami operator are of crucial importance when it comes to understanding the geometry of the manifold, as they are a natural basis for decomposing functions and they encode geometric and physical properties.
Some properties of this spectrum are the following:

- Non-negativity: if the manifold $\mathcal{M}$ is compact, the spectrum consists of a discrete sequence of non-negative eigenvalues. As already mentioned in the previous sections, it is also necessary to define the operator as $\Delta f = -div(\nabla f)$ to ensure non-negativity. Thus the eigenvalues will form a sequence of the type:

  $$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$$

  Here the first eigenvalue $\lambda_0 = 0$ corresponds to constant eigenfunctions.

- Eigenfunction Basis: The eigenfunctions form an orthonormal basis for $L^2(\mathcal{M})$ (square integrable functions on the manifold).

- Dependence on Geometry: the spectrum encodes geometric information about the manifold (volume, curvature, topology, ...), however distinct manifolds can have the same spectrum. These are called isospectral manifolds.

The spectrum of the Laplace-Beltrami operator allows for both physical and geometrical interpretations.

As for the geometric interpretation, the spectrum of Δ also connects to the curvature of the manifold. Even though we do not know much about this relation, there is an important result in [12] the authors have explored the properties of the heat kernel associated to the Schrödinger operator on complete Riemannian manifolds. One of the results is the Li-Yau gradient estimates, that provide upper and lower bounds to positive solutions of the heat equations using the Ricci curvature.

In particular, Li and Yau gradient estimate for positive solutions $u(x, t)$ on a Riemannian manifold with Ricci curvature bounded below states:

$$\frac{|\nabla u|^2}{u} - \alpha \frac{\partial u}{\partial t} \leq \frac{n\alpha^2}{2t}$$

Here α is a positive constant that can be chosen at convenience (although there are various methods to optimize its value).

In the same paper they also provide a lower bound for the first non-zero eigenvalue in terms of curvature and diameter of the manifold:

$$\lambda_1 \geq \frac{C(n)}{diam(\mathcal{M}) - C'(n)K}$$

Where K is the lower bound on Ricci curvature, and $diam(\mathcal{M})$ is the diameter of the manifold. The result shows that as curvature increases, the spectrum of the Laplace-Beltrami operator also shifts upwards. Thus manifolds with highly positive curvature have higher fundamental frequencies.

As for the physical interpretation, in *Can one hear the shape of a drum?* [11] the author studies the role of eigenvalues and eigenfunctions of Δ in the context of vibrating membranes. The main question is whether the spectrum of the Laplace-Beltrami operator uniquely determines the geometry of the manifold. The connection to physical acoustics arises from the fact that, if $\mathcal{M}$ represents a drumhead, then eigenfunctions describe standing wave patterns of vibration, while eigenvalues are squared frequencies of oscillation.

Examples of Spectrum

On the n-dimensional sphere S^n, with standard metric, the eigenvalues are $\lambda_\ell = \ell(\ell + n - 1), \quad \ell \in \mathbb{N}$, while the eigenfunctions are spherical harmoinics $Y_{\ell,m}(\theta, \phi) = P_\ell^m(\cos\theta)e^{im\phi}$.

To reach this result, let's define S^n as:

$$S^n = \{x \in \mathbb{R}^{n+1} \mid |x|^2 = 1\}.$$

The Laplace–Beltrami operator Δ_{S^n} on the sphere is the restriction of the Laplacian in $\mathbb{R}^{n+1}$ to functions that are invariant under the constraint $|x| = 1$. In spherical coordinates, it is given by:

$$\Delta_{S^n} = \frac{1}{\sin^{n-1}\theta}\frac{\partial}{\partial\theta}\left(\sin^{n-1}\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\Delta_{S^{n-1}}.$$

For the case of S^2 (the standard 2-sphere), this simplifies to:

$$\Delta_{S^2} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}.$$

Now, we seek eigenfunctions $Y(\theta, \phi)$ satisfying:

$$\Delta_{S^n} Y = \lambda Y.$$

Using separation of variables, the eigenfunctions take the form:

$$Y(\theta, \phi) = P(\theta)e^{im\phi},$$

where $e^{im\phi}$ are Fourier modes with integer m. The function $P(\theta)$ satisfies the associated Legendre equation, leading to spherical harmonics:

$$Y_{\ell,m}(\theta, \phi) = P_\ell^m(\cos\theta)e^{im\phi}.$$

The corresponding eigenvalues are:

$$\lambda_\ell = \ell(\ell + n - 1), \quad \ell \in \mathbb{N}.$$

2 Graph Laplacian and Manifold Learning

The manifold Laplacian is one of the key objects associated to a general differentiable Riemannian manifold. In fact, several manifold learning algorithms are closely related to the Laplacian. However in most manifold learning settings the underlying manifold is usually unknown, hence functional maps from the manifold have to be estimated using point cloud data. The common approximation strategy in these methods is to construct an adjacency graph associated to a point cloud. Since the graph is a proxy for the manifold, inference based on the structure of the graph should corresponds to the desired inference based on the geometric structure of the manifold.

2.1 Graph Laplacian

A weighted graph $G = (V, E)$ is a set of vertices $v_1, v_2, ...v_n \in V$ and weighted edges connecting these vertices represented by an adjacency matrix W. W is a symmetric matrix with nonnegative entries. The Laplacian matrix of a weighted graph G is the matrix $L = D - W$ where D is a diagonal matrix $D(i,i) = \sum_j W(i,j)$.
Given a set of points $\mathcal{S}_n = \{\mathbf{x}_1, ..., \mathbf{x}_n\} \in \mathbb{R}^k$, we construct a graph G whose vertices are data points. We put $W_n^t(i,j) = e^{-\frac{||\mathbf{x}_i - \mathbf{x_j}||^2}{4t}}$. We will denote the corresponding graph Laplacian by $\hat{L}_n^t = D_n^t - W_n^t$. Note that we suppress the dependence on $\mathcal{S}_n$ to simplify the notation. $\hat{L}_n^t$ can be then thought as an operator on functions, defined on the graph of data points. If $f : V \to \mathbb{R}$, then:

$$\hat{L}_n^t f(\mathbf{x_i}) = f(\mathbf{x_i}) \sum_j e^{-\frac{||\mathbf{x}_i - \mathbf{x}_j||^2}{4t}} - \sum_j f(\mathbf{x_j}) e^{-\frac{||\mathbf{x}_i - \mathbf{x}_j||^2}{4t}}$$

3 Point-wise convergence

In the paper *Towards a Theoretical Foundation for Laplacian-Based Manifold Methods* [3] the authors Belkin and Niyogi show that the graph Laplacian of a point cloud converges to the Laplace-Beltrami operator of the underlying manifold.
Consider a compact k-dimensional differentiable manifold $\mathcal{M}$ isometrically embedded in $\mathbb{R}^N$. It is assumed that data is sampled from a uniform distribution in the induced measure on $\mathcal{M}$.
Given the data $\mathcal{S}_n = \{\mathbf{x}_1, ..., \mathbf{x}_n\} \in \mathbb{R}^N$ sampled i.i.d. from this probability distribution, the associated Laplacian operator $\hat{\mathbf{L}}_n^t$ can be constructed. The above mentioned paper shows that for a fixed function $f \in C\infty(\mathcal{M})$ and for a fixed point $\mathbf{p} \in \mathcal{M}$ after appropriate scaling the operator $\hat{\mathbf{L}}_n^t$ converges to the true Laplace-Beltrami operator on the manifold. In particular the uniform distribution on the manifold can arise as the Riemann structure on $\mathcal{M}$ induces a measure corresponding to the volume form, which will be denoted μ. For a compact manifold $\mathcal{M}$ total volume of $\mathcal{M}$ is guaranteed to be finite, allowing us to define the canonical uniform probability distribution on $\mathcal{M}$.

Theorem. Let the data points $\mathbf{x}_1, ..., \mathbf{x}_n$ be sampled from a uniform distribution on a manifold $\mathcal{M} \subset \mathbb{R}^N$. Let $t_n = n^{-\frac{1}{k+2+\alpha}}$ where $\alpha > 0$ and let $f \in C^\infty(\mathcal{M})$. Then there is a constant C such that in probability,

$$\lim_{n\to\infty} C\frac{(4\pi t_n)^{-\frac{k+2}{2}}}{n}\hat{\mathbf{L}}_n^{t_n} f(\mathbf{x}) = \Delta_{\mathcal{M}} f(\mathbf{x})$$

Another stronger (uniform) version of this theorem is the following:

Theorem. Let the data points $\mathbf{x}_1, ..., \mathbf{x}_n$ be sampled from a uniform distribution on a manifold $\mathcal{M} \subset \mathbb{R}^N$. Let $\mathcal{F}$ be the space of functions $f \in C^\infty(\mathcal{M})$ such that Δf is Lipschitz with Lipschitz constant equal to 1. Then there exists a sequence of real numbers $t_n \to 0$ and a constant C such that in probability:

$$\lim_{n\to\infty} \sup_{\mathbf{x}\in\mathcal{M}, f\in\mathcal{F}} \left| C\frac{(4\pi t_n)^{-\frac{k+2}{2}}}{n}\hat{\mathbf{L}}_n^{t_n} f(\mathbf{x}) - \Delta_{\mathcal{M}} f(\mathbf{x}) \right| = 0$$

3.1 Proof of Point-wise Convergence

The proof will show that the Laplace Beltrami operator can be estimated using the following quantities:

$$\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{x}||^2}{4t}} f(\mathbf{x}) d\mu_{\mathbf{x}}$$

$$f(\mathbf{p}) \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{x}||^2}{4t}} d\mu_{\mathbf{x}}$$

Because in fact, these two quantities can be empirically estimated from the point cloud. Estimating the Laplace-Beltrami operator will provide a connection to $\mathbf{L}_n^t$.

Lemma. Given any open set $\mathcal{B} \subset \mathcal{M}$, $\mathbf{p} \in \mathcal{B}$, for any $l \in N$,

$$\int_{\mathcal{B}\in\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mu_{\mathbf{y}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mu_{\mathbf{y}} = o(t^l)$$

as $t \to 0$.

Proof. Let $d = inf_{\mathbf{x}\notin\mathcal{B}} ||\mathbf{p}-\mathbf{x}||^2$ and let M be the measure of the complement to $\mathcal{B}$ ($M = \mu(\mathcal{M}-\mathcal{B})$). Since $\mathcal{B}$ is open and $\mathcal{M}$ is locally compact, then $d > 0$. Hence:

$$\left| \int_{\mathcal{B}\in\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mu_{\mathbf{y}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mu_{\mathbf{y}} \right| \leq M \sup_{x\in\mathcal{M}} (|f(\mathbf{x}|) e^{-\frac{d^2}{4t}}$$

Where M and $\sup_{x\in\mathcal{M}}(|f(\mathbf{x}|)$ are constant, while $e^{-\frac{d^2}{4t}}$ approaches 0 faster than any polynomial as $t \to 0$. Hence the lemma. □

Given the lemma, it is now possible to replace the integral over the manifold with an integral over a small open set around $\mathbf{p}$. This will allow for a change of variables to standard geodesic coordinates through $\mathbf{y} = exp_{\mathbf{p}}(\mathbf{x})$.

Given a function $f : \mathcal{M} \to \mathbb{R}$, we denote it, in geodesic coordinates, by:

$$\tilde{f}(\mathbf{x}) = f(exp(\mathbf{x}))$$

The following statement relates the Laplace-Beltrami operator and the Euclidean Laplacian. It is presented without proof, but the proof can be found in [9].

Lemma.

$$\Delta_{\mathcal{M}} f(\mathbf{p}) = \Delta_{\mathbb{R}^k} \tilde{f(0)}$$

Now, since $exp_{\mathbf{p}} : T\mathcal{M}_{\mathbf{p}} = \mathbb{R}^k \to \mathcal{M}$ is locally invertible, we can choose an open $\tilde{\mathcal{B}} \subset \mathbb{R}^k$ such that $exp_{\mathbf{p}}$ is a diffeomorphism onto its image $\mathcal{B} \subset \mathcal{M}$. This will allow to reduce the Laplace-Beltrami operator to the Laplace operator on $\mathbb{R}^k$, which is easier to analyze.

Lemma. The following change of variable formula holds:

$$\int_{\mathcal{B}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mu_{\mathbf{y}} = \int_{\tilde{\mathcal{B}}} e^{-\frac{\phi(x)}{4t}} \tilde{f}(\mathbf{x}) det(d\, exp(\mathbf{x})) d\mathbf{x}$$

where $\phi(\mathbf{x})$ is a function such that $\phi(\mathbf{x}) = ||\mathbf{x}||^2 + O(||\mathbf{x}||^4)$.

Proof. The result is gotten by applying the usual change of variable formula for manifold integrals and using the relation between geodesic and chordal distance. □

Lemma:

$$|\Delta_{\mathbb{R}^k}(det(d\, exp_{\mathbf{p}})(\mathbf{x})| = \frac{scal(\mathbf{p})}{3} + O(||\mathbf{x}||)$$

Proof. Let's consider the determinant of the differential of the exponential map $d\exp_{\mathbf{p}}$. For small $\mathbf{x}$, the determinant has the following Taylor expansion (proof is in [10]):

$$\det(d\exp_{\mathbf{p}})(\mathbf{x}) = 1 - \frac{1}{6}\operatorname{scal}(\mathbf{p})|\mathbf{x}|^2 + O(|\mathbf{x}|^3),$$

where $scal(\mathbf{p})$ is the scalar curvature at $\mathbf{p}$. This result follows from properties of Jacobi fields (a vector field along a geodesic in a Riemannian manifold that describes how nearby geodesics spread or converge). Specifically, the curvature at $\mathbf{p}$ affects the behavior of the exponential map.

Now, we will consider the Laplacian of this function $\det(d\exp_{\mathbf{p}})(\mathbf{x})$ and apply it in a neighborhood of $\mathbf{p}$. Since the function is radially symmetric, it depends only on the distance $r = |\mathbf{x}|$ from $\mathbf{p}$. The Laplacian in $\mathbb{R}^k$ for such a radially symmetric function $f(r)$ is given by:

$$\Delta_{\mathbb{R}^k} f(r) = \frac{d^2}{dr^2} f(r) + \frac{k-1}{r} \frac{d}{dr} f(r).$$

For $f(r) = 1 - \frac{1}{6}\operatorname{scal}(\mathbf{p})r^2 + O(r^3)$, we compute the derivatives:

$$\frac{d}{dr}\left(1 - \frac{1}{6}\operatorname{scal}(\mathbf{p})r^2\right) = -\frac{1}{3}\operatorname{scal}(\mathbf{p})r + O(r^2).$$

$$\frac{d^2}{dr^2}\left(1 - \frac{1}{6}\operatorname{scal}(\mathbf{p})r^2\right) = -\frac{1}{3}\operatorname{scal}(\mathbf{p}) + O(r).$$

Using the Laplacian formula:

$$\Delta_{\mathbb{R}^k}\left(1-\frac{1}{6}\operatorname{scal}(\mathbf{p})r^2+O(r^3)\right)=\frac{d^2}{dr^2}\left(1-\frac{1}{6}\operatorname{scal}(\mathbf{p})r^2\right)+\frac{k-1}{r}\frac{d}{dr}\left(1-\frac{1}{6}\operatorname{scal}(\mathbf{p})r^2\right).$$

Evaluating the terms:

- The first derivative term gives: $\frac{k-1}{r}\left(-\frac{1}{3}\operatorname{scal}(\mathbf{p})r\right)=-\frac{1}{3}(k-1)\operatorname{scal}(\mathbf{p})+O(r)$.
- The second derivative term gives: $-\frac{1}{3}\operatorname{scal}(\mathbf{p})$.

Thus, combining these terms, we obtain the Laplacian:

$$\Delta_{\mathbb{R}^k}\left(1-\frac{1}{6}\operatorname{scal}(\mathbf{p})r^2+O(r^3)\right)=-\frac{k}{3}\operatorname{scal}(\mathbf{p})+O(r).$$

This establishes the following result:

$$\Delta_{\mathbb{R}^k}\det(d\exp_{\mathbf{p}})(\mathbf{x})=-\frac{k}{3}\operatorname{scal}(\mathbf{p})+O(|\mathbf{x}|).$$

□

Now, recalling that k is the dimension of the manifold, and $scal(\mathbf{p})$ is the scalar curvature of the manifold at $\mathbf{p}$, we state the following lemma:

Lemma. There exists a constant C such that:

$$\frac{\partial}{\partial t}\Big((4\pi t)^{-k/2}\int_{\mathcal{B}}e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}}f(\mathbf{y})d\mu_{\mathbf{y}}\Big)\Big|_0=\Delta_{\mathcal{M}}f(\mathbf{p})+\frac{1}{3}k\ scal(\mathbf{p})f(\mathbf{p})+Cf(\mathbf{p})$$

Proof. First, by the previous lemmas we can write the following equality:

$$\frac{\partial}{\partial t}\Big((4\pi t)^{-k/2}\int_{\mathcal{B}}e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}}f(\mathbf{y})d\mu_{\mathbf{y}}\Big)\Big|_0=\frac{\partial}{\partial t}\Big((4\pi t)^{-k/2}\int_{\tilde{\mathcal{B}}}e^{-\frac{\phi(x)}{4t}}\tilde{f}(\mathbf{x})det(d\,exp(\mathbf{x}))d\mathbf{x}\Big)\Big|_0$$

Let us now state the following result:

Let $\varphi:\mathbb{R}^k\to\mathbb{R}^k$ be a differentiable function such that $\varphi(x)=x+O(x^3)$ i.e. the Taylor expansion does not have any terms of degree 2 at the origin. Then for any open set $\tilde{\mathcal{B}}$ containing the origin the following two expression hold:

$$f(0)=\lim_{t\to 0}(4\pi t)^{-n/2}\int_{\tilde{\mathcal{B}}\subset\mathbb{R}^k}e^{-\frac{\varphi(y)^2}{4t}}f(y)dy$$

$$\Delta f(0) = -\frac{\partial}{\partial t}\Big((4\pi t)^{-n/2}\int_{\tilde{\mathcal{B}}\subset\mathbb{R}^k} e^{-\frac{\varphi(y)^2}{4t}} f(y)dy\Big)\Big|_0 + Cf(0)$$

Where C is a constant depending only on φ.

Proof for the previous statement can be found in [3] in *Auxiliary and Technical Lemmas.*

We can use the previous result to write the following equality:

$$\frac{\partial}{\partial t}\Big((4\pi t)^{-k/2}\int_{\tilde{\mathcal{B}}} e^{-\frac{\phi(x)}{4t}}\tilde{f}(\mathbf{x})det(d\,exp(\mathbf{x}))d\mathbf{x}\Big)\Big|_0 = \Delta_{\mathbb{R}^k}(\tilde{f}det(d\,exp_{\mathbf{p}})(0) + C\tilde{f}(0)$$

We can now apply the previous result for exponential maps:

$$|\Delta_{\mathbb{R}^k}(det(d\,exp_{\mathbf{p}})(\mathbf{x})| = \frac{scal(\mathbf{p})}{3} + O(||\mathbf{x}||)$$

Next, we apply the Laplacian to the function $\tilde{f}(\mathbf{x})\det(d\exp_{\mathbf{p}})(\mathbf{x})$, where $\tilde{f}$ is a smooth function with $\tilde{f}(\mathbf{0}) = f(\mathbf{p})$. Using the standard Laplacian product rule:

$$\Delta_{\mathbb{R}^k}(\tilde{f}\det(d\exp_{\mathbf{p}})) = \tilde{f}\Delta_{\mathbb{R}^k}\det(d\exp_{\mathbf{p}}) + 2\nabla\tilde{f}\cdot\nabla\det(d\exp_{\mathbf{p}}) + \det(d\exp_{\mathbf{p}})\Delta_{\mathbb{R}^k}\tilde{f}.$$

Since $\det(d\exp_{\mathbf{p}})(\mathbf{x})$ has no linear terms in its Taylor expansion at 0 and is radially symmetric, we can compute the Laplacian of $\det(d\exp_{\mathbf{p}})$ at 0:

$$\Delta_{\mathbb{R}^k}\det(d\exp_{\mathbf{p}})(\mathbf{0}) = -\frac{k}{3}\operatorname{scal}(\mathbf{p}).$$

Thus, the Laplacian of the product is:

$$\Delta_{\mathbb{R}^k}(\tilde{f}\det(d\exp_{\mathbf{p}}))(0) = \Delta_{\mathbb{R}^k}\tilde{f}(0) + \frac{1}{3}k\operatorname{scal}(\mathbf{p})f(\mathbf{p}).$$

Finally, recalling that $\Delta_{\mathbb{R}^k}\tilde{f}(0) = \Delta_{\mathcal{M}}f(\mathbf{p})$ for a function f on the manifold, we obtain the result:

$$\Delta_{\mathbb{R}^k}(\tilde{f}\det(d\exp_{\mathbf{p}}))(0) = \Delta_{\mathcal{M}}f(\mathbf{p}) + \frac{1}{3}k\operatorname{scal}(\mathbf{p})f(\mathbf{p}).$$

Thus, combining the properties of the Laplacian and the Taylor expansion of $\det(d\exp_{\mathbf{p}})$, we have derived the desired result. □

Lemma.

$$\lim_{t\to 0}(4\pi t)^{-\frac{k+2}{2}}\left(\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}}f(\mathbf{p})d\mu_{\mathbf{y}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}}f(\mathbf{y})d\mu_{\mathbf{y}}\right) = \Delta_{\mathcal{M}}f(\mathbf{p})$$

Proof. Consider the constant function $g(\mathbf{y}) = f(\mathbf{p})$. By applying the previous lemma to this function we obtain:

$$\frac{\partial}{\partial t}\Big((4\pi t)^{-k/2}\int_{\mathcal{B}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{p})d\mu_{\mathbf{y}}\Big)\Big|_0 = \frac{1}{3}k\ scal(\mathbf{p})f(\mathbf{p}) + Cf(\mathbf{p})$$

Now, let:

$$A(t) = (4\pi t)^{-\frac{k+2}{2}}\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y})d\mu_{\mathbf{y}}$$

Recall that the heat kernel had the property that: $f(\mathbf{x}) = \lim_{t\to 0}\mathbf{H}^t f(\mathbf{x})$, we can see that:

$$A(0) = \lim_{t\to 0}\ (4\pi t)^{-\frac{k}{2}}\int_{\mathcal{B}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{p})d\mu_{\mathbf{y}} = f(\mathbf{p})$$

Thus, by the definition of derivative and the previous lemma, we get that:

$$\Delta_{\mathcal{M}} f(\mathbf{p}) = \lim_{t\to 0}\frac{A(t)-A(0)}{t} =$$

$$\lim_{t\to 0}\left(\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{p})\mathbf{d}\mu_{\mathbf{y}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y})\mathbf{d}\mu_{\mathbf{y}}\right)$$

□

Theorem. Let data points $\mathbf{x}_1, ..., \mathbf{x}_n$ be sampled from an i.i.d. fashion from a uniform distribution on a compact submanifold $\mathcal{M} \subset \mathbb{R}^N$. Fix $\mathbf{p} \in \mathcal{M}$. Let $\hat{\mathbf{L}}_n^{t_n}$ be the associated operator. Put $t_n = n^{-\frac{1}{k+2+\alpha}}$ where $\alpha > 0$, $\alpha \in \mathbb{R}$. Then in probability:

$$\lim_{n\to\infty}(4\pi t_n)^{-\frac{k+2}{2}}\hat{\mathbf{L}}_n^{t_n} f(\mathbf{x}) = \frac{\Delta_{\mathcal{M}} f(\mathbf{p})}{vol(\mathcal{M})}$$

Proof. Recall that the extension of the graph Laplacian $\hat{\mathbf{L}}_n^t$ applied to f at $\mathbf{p}$ is

$$\hat{\mathbf{L}}_n^t f(\mathbf{p}) = \frac{1}{n}\left(\sum_{i=1}^{n} e^{-\frac{||\mathbf{p}-\mathbf{x_i}||^2}{4t}} f(\mathbf{p}) - \sum_{i=1}^{n} e^{-\frac{||\mathbf{p}-\mathbf{x_i}||^2}{4t}} f(\mathbf{x}_i)\right)$$

Observe that $\hat{\mathbf{L}}_n^t f(\mathbf{p})$ is the empirical average of n independent random variables with expectation:

$$\mathbb{E}\hat{\mathbf{L}}_n^t f(\mathbf{p}) = \left(f(\mathbf{p})\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} d\mu_{\mathbf{y}} - \int_{\mathcal{M}} f(\mathbf{y})e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} d\mu_{\mathbf{y}}\right)$$

We are now stating the following theorem, proof can be found in [3].

Theorem (Hoeffding). Let $X_1, ..., X_n$ be i.i.d. random variables, such that $|X_i| \leq K$. Then:

$$\mathbb{P}\Big\{\Big|\frac{\sum_i X_i}{n} - \mathbb{E}X_i\Big| > \varepsilon\Big\} < 2e^{-\frac{\varepsilon^2 n}{2K^2}}$$

By applying this theorem to the graph Laplacian:

$$\mathbb{P}[(4\pi t)^{-\frac{k+2}{2}}|\hat{\mathbf{L}}_n^t f(\mathbf{p}) - \mathbb{E}\hat{\mathbf{L}}_n^t f(\mathbf{p})| > \varepsilon] \leq e^{-\varepsilon^2 n(4\pi t)^{k+2}}$$

Let t be a function of n, and in particular $t = t_n = \frac{1}{n}^{\frac{1}{k+2+\alpha}}$ with $\alpha > 0$, we see that for any fixed $\varepsilon > 0$:

$$\lim_{n\to\infty} \mathbb{P}[(4\pi t_n)^{-\frac{k+2}{2}}|\hat{\mathbf{L}}_n^{t_n} f(\mathbf{p}) - \mathbb{E}\hat{\mathbf{L}}_n^{t_n} f(\mathbf{p})| > \varepsilon] = 0$$

Thus, recalling :

$$\lim_{t\to 0} (4\pi t)^{-\frac{k+2}{2}} \left(f(\mathbf{p}) \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} d\mu_{\mathbf{y}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} f(\mathbf{y}) d\mu_{\mathbf{y}} \right) = \Delta_{\mathcal{M}} f(\mathbf{p})$$

and

$$\mathbb{E}\hat{\mathbf{L}}_n^t f(\mathbf{p}) = \left(f(\mathbf{p}) \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} d\mu_{\mathbf{y}} - \int_{\mathcal{M}} f(\mathbf{y}) e^{-\frac{||\mathbf{p}-\mathbf{y}||^2}{4t}} d\mu_{\mathbf{y}} \right)$$

we can observe that, as desired:

$$\lim_{n\to\infty} (4\pi t_n)^{-\frac{k+2}{2}} \hat{\mathbf{L}}_n^{t_n} f(\mathbf{x}) = \lim_{n\to\infty} (4\pi t_n)^{-\frac{k+2}{2}} \mathbb{E}\hat{\mathbf{L}}_n^{t_n} f(\mathbf{x}) = \frac{\Delta_{\mathcal{M}} f(\mathbf{p})}{vol(\mathcal{M})}$$

□

4 Spectral Convergence

In *Convergence of Laplacian Eigenmaps* [2], the authors Belkin and Niyogi show the convergence of the eigenvectors of the graph Laplacian to the eigenfunctions associated to a point cloud dataset to eigenfunctions of the Laplace-Beltrami operator when the data is sampled from a uniform probability distribution on an embedded manifold.
Notice that, while the Laplace-Beltrami operator has infinitely many eigenfunctions, the graph Laplacian (being a matrix) has only a finite number of eigenvectors. Indeed the convergence will be proven for the i-th smallest eigenvalues. Then in most of manifold learning application only the eigenvectors relative to the smallest eigenvalues are used.
It is assumed that the manifold $\mathcal{M}$ is a compact infinitely differentiable Riemann submanifold of $\mathbb{R}^N$ without boundary. We will denote by λ_i the i-th smallest eigenvalue of the Laplace-Beltrami operator Δ and by e_i the relative eigenfunction.

As in many other examples, in here we will use the definition of Laplace-Beltrami operator that is:

$$\Delta f = -div(\nabla f)$$

So that the Laplace-Beltrami operator Δ is a positive semi-definite operator and we can work with non-negative eigenvalues.

Let us define the operator $\mathbf{L}^t : L^2(\mathcal{M}) \to L^2(\mathcal{M})$, where $L^2(\mathcal{M})$ is the space of square integrable functions:

$$\mathbf{L}^t f(\mathbf{p}) = (4\pi t)^{-\frac{k+2}{2}} \Big(\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{q}||^2}{4t}} f(\mathbf{p}) d\mu_{\mathbf{q}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{q}||^2}{4t}} f(\mathbf{q}) d\mu_{\mathbf{q}} \Big)$$

Recalling that μ is the standard measure on the manifold.
Given our data-points $\mathbf{x}_i$, the empirical version of this operator is given by:

$$\hat{\mathbf{L}}_n^t f(\mathbf{p}) = \frac{(4\pi t)^{-\frac{k+2}{2}}}{n} \Big(\sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}} f(\mathbf{p}) - \sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}} f(\mathbf{x}_i) \Big)$$

Later, in the Laplacian Eigenmaps algorithm, it will be clear that the operator $\hat{\mathbf{L}}_n^t f(\mathbf{p})$ is the point cloud Laplacian that forms a basis of the Laplacian Eigenmaps algorithm for manifold learning.

We will show that there is a way to choose the sequence t_n in such a way that eigenfunctions of the empirical operator $\hat{\mathbf{L}}_n^{t_n} f(\mathbf{p})$ converge to eigenfunctions of the Laplace-Beltrami operator in probability (as the points $\mathbf{x}_i$ are randomly i.i.d. sampled from $\mathcal{M}$).

Thus, spectral convergence is stated in the following theorem:

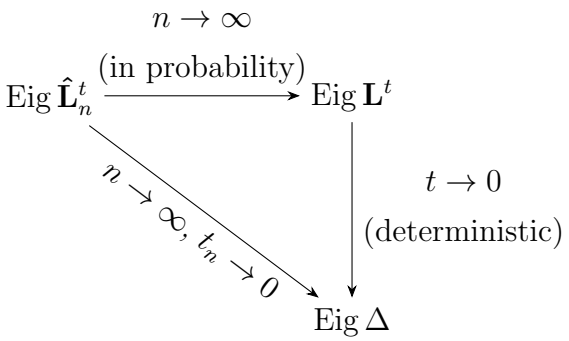
Theorem. Let $\lambda_{n,i}^t$ be the i-th eigenvalue of $\hat{\mathbf{L}}_n^t$ and $e_{n,i}^t$ be the corresponding eigenfunction (which for each fixed i will be shown to exist for t sufficiently small). Let λ_i and e_i be the corresponding eigenvalue and eigenfunction of Δ respectively. Then there exists a sequence $t_n \to 0$, such that

$$\lim_{n\to\infty} \lambda_{n,i}^{t_n} = \lambda_i$$
$$\lim_{n\to\infty} ||e_{n,i}^{t_n}(x) - e_i(x)||_2 = 0$$

where the limits are in probability.

4.1 Proof of Spectral Convergence

We are first going to prove the spectral convergence of the functional approximation $\mathbf{L}^t$ to Δ as $t \to 0$ and then the spectral convergence of the empirical approximation $\hat{\mathbf{L}}_n^t$ to $\mathbf{L}^t$ as the number of data-points n tends to infinity.



This will be shown by the following two Theorems:

Theorem (Spectral Convergence of Functional Approximation). Let λ_i, λ_i^t, e_i, e_i^t be the i-th smallest eigenvalue and the corresponding eigenfunction of Δ and $\mathbf{L}^t$ respectively. Then:

$$\lim_{t\to 0} |\lambda_i - \lambda_i^t| = 0$$
$$\lim_{t\to 0} ||e_i - e_i^t||_2 = 0$$

Theorem (Spectral Convergence of Empirical Approximation). For a fixed sufficiently small t, let $\lambda_{n,i}^t$ and λ_i^t be the i-th eigenvalue of $\hat{\mathbf{L}}_n^t$ and $\mathbf{L}^t$ respectively. Let $e_{n,i}^t$ and e_i^t be the corresponding eigenfunctions. Then:

$$\lim_{t\to 0}\lambda_{n,i}^t = \lambda_i^t$$

$$\lim_{t\to 0}||e_{n,i}^t - e_i^t||_2 = 0$$

assuming that $\lambda_i^t \leq \frac{1}{2t}$. The convergence is almost sure.

Then the spectral convergence of $\hat{\mathbf{L}}_n^t$ to Δ follows from these two theorems, that imply the convergence $\mathbf{Eig}\ \hat{\mathbf{L}}_n^{\,t} \xrightarrow{n\to\infty} \mathbf{Eig}\ \mathbf{L}^t$ and $\mathbf{Eig}\ \mathbf{L}^t \xrightarrow{t\to 0} \mathbf{Eig}\,\Delta$.
$\mathbf{Eig}\ \hat{\mathbf{L}}_n^{\,t} \xrightarrow{n\to\infty} \mathbf{Eig}\ \mathbf{L}^t$ is almost sure for $\lambda_i \leq \frac{1}{2}t^{-1}$. Given any $i \in \mathbb{N}$ and $\varepsilon > 0$, we can choose $t' < 2\lambda_i^{-1}$ such that for all $t < t'$ we have $||e_i - e_i^t||_2 < \varepsilon/2$.
On the other hand, by using the first arrow we see that:

$$\lim_{n\to\infty}\mathbb{P}\Big\{||e_{n,i}^t - e_i^t||_2 \geq \frac{\varepsilon}{2}\Big\}$$

Thus if for any $p > 0$ and for each t there exists and N such that:

$$\mathbb{P}\{||e_{n,i}^t - e_i^t||_2 > \varepsilon\} < p$$

Thus we can see that for any N and for any probability $p(N)$ there exists a t_N such that:

$$\forall n > N \qquad \{||e_{n,i}^{t_N} - e_i||_2 > \varepsilon\} < p(N)$$

Letting $p(N)$ tend to zero we obtain convergence in probability.

4.1.1 Proof: Spectral Convergence of Functional Approximation

The manifold $\mathcal{M}$ is assumed to be compact, smooth, k-dimensional embedded in $\mathbb{R}^N$. We define the operator

$$\mathbf{L}^t f(\mathbf{p}) = (4\pi t)^{-\frac{k+2}{2}} \Big(\int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{q}||^2}{4t}} f(\mathbf{p}) d\mu_{\mathbf{q}} - \int_{\mathcal{M}} e^{-\frac{||\mathbf{p}-\mathbf{q}||^2}{4t}} f(\mathbf{q}) d\mu_{\mathbf{q}} \Big)$$

We want to show in what sense will this converge to the Laplace-Beltrami operator Δ. Let us first recall the heat equation (with a negative sign as we have defined $\Delta f = -div(\nabla)$:

$$\Delta h(x,t) = -\frac{\partial h(x,t)}{\partial t}$$

where $h(x,t)$ is the heat at time t and position x. Let the initial heat distribution be $f(x) = h(x,0)$. Then, by definition of derivative:

$$\Delta f = \Delta h(x,0) = -\frac{\partial h(x,t)}{\partial t}\bigg|_{t=0} = -\lim_{t\to 0}\frac{1}{t}(h(x,t) - f(x))$$

As stated in Section 1.2, the solution to the heat equation at time t can be written as:

$$\mathbf{H}^t f(x) = h(x,t) = \int_{\mathcal{M}} f(y) H^t(x,y) d\mu_y$$

Observe that we can rewrite the Laplace Beltrami operator in terms of the heat operator:

$$\Delta f = -\lim_{t\to 0}\frac{1}{t}(\mathbf{H}^t f(x) - f(x)) = \lim_{t\to 0}\frac{1}{t}(f(x) - \mathbf{H}^t f(x))$$

Thus we can write the Δ operator as:

$$\Delta = \lim_{t\to 0}\frac{\mathbf{1} - \mathbf{H^t}}{t}$$

This result could be have also been derived from the fact that, as explained in the previous sections, it is possible to write $\mathbf{H}^t = e^{-\Delta t}$ and through a Taylor expansion we reach the same result.

Observe that, given λ an eigenvalue of Δ, and f its relative eigenfunction ($\Delta f = \lambda f$), we have that :

$$\mathbf{H}^t f = e^{-\Delta t} f = e^{-\lambda t} f$$

Thus if f is an eigenfunction of Δ, it is also an eigenfunction of $\mathbf{H}^t$ and in particular if λ is an eigenvalue of Δ, then $e^{-\lambda t}$ is an eigenvalue of $\mathbf{H}^t$

Thus also eigenfunctions of $\frac{\mathbf{1}-\mathbf{H^t}}{t}$ coincide with the eigenfunctions of the Laplace operator Δ, as:

$$\frac{\mathbf{1} - \mathbf{H^t}}{t} f = \frac{f - \mathbf{H^t} f}{t} = \frac{f - e^{-\lambda t} f}{t} = \frac{1 - e^{-\lambda t}}{t} f$$

Hence, eigenfunctions of $\frac{\mathbf{1}-\mathbf{H^t}}{t}$ coincide with those of Δ, and given λ_i the i-th eigenvalue of Δ, we have that $\frac{1-e^{-\lambda_i t}}{t}$ is the i-th eigenvalue of $\frac{\mathbf{1}-\mathbf{H^t}}{t}$.

To prove spectral convergence of the functional approximation, we claim that $\mathbf{L}^t$ is a relatively bounded perturbation of $\frac{\mathbf{1}-\mathbf{H^t}}{t}$. This will imply convergence of eigenfunctions of $\mathbf{L}^t$ to those of $\frac{\mathbf{1}-\mathbf{H^t}}{t}$.

Definition (Relatively Bounded Perturbation): Let A be a densely defined linear operator on a Hilbert space and B a second operator. We say that B is a relatively bounded perturbation of A if there exist constants $a, b \geq 0$ such that for all f in the domain of A,

$$\|Bf\| \leq a\|Af\| + b\|f\|.$$

If a can be made arbitrarily small (e.g., $a \to 0$ as some parameter $\to 0$), B is said to be relatively compact or negligible compared to A in the appropriate limit.

Let the perturbation operator $\mathbf{R}^t$ be defined as:

$$\mathbf{R}^t = \frac{\mathbf{1} - \mathbf{H^t}}{t} - \mathbf{L}^t.$$

Theorem. For any $0 < \varepsilon < \frac{2}{k+2}$, there exists a constant C such that for all sufficiently small t,

$$\frac{|\langle \mathbf{R}^t f, f\rangle|}{\left\langle \frac{\mathbf{1}-\mathbf{H^t}}{t} f, f\right\rangle} \leq C \max\left(t^{\frac{2}{k+2}-\varepsilon}, t^{\frac{2}{k+2}\varepsilon}\right).$$

In particular,

$$\lim_{t\to 0} \sup_{\|f\|_2=1} \frac{\langle \mathbf{R}^t f, f\rangle}{\left\langle \frac{\mathbf{1}-\mathbf{H^t}}{t} f, f\right\rangle} = 0,$$

and hence $\mathbf{R}^t$ is dominated by $\frac{\mathbf{1}-\mathbf{H^t}}{t}$ on L^2 as $t \to 0$.
Proof of this theorem is in [2].

This result shows that $\mathbf{L}^t$ differs from $\frac{\mathbf{1}-\mathbf{H^t}}{t}$ by a perturbation that vanishes in operator norm relative to $\frac{\mathbf{1}-\mathbf{H^t}}{t}$ as $t \to 0$. Therefore, by classical perturbation theory, the eigenvalues and eigenfunctions of $\mathbf{L}^t$ converge to those of $\frac{\mathbf{1}-\mathbf{H^t}}{t}$, which themselves converge to those of the Laplacian Δ. Thus, we have established spectral convergence of the functional approximation of the Laplacian.

4.1.2 Proof: Spectral Convergence of Empirical Approximation

Proposition For t small enough

$$SpecEss(\mathbf{L}^t) \subset (\frac{1}{2}t^{-1}, \infty)$$

Where $SpecEss$ is the essential spectrum of the operator (the essential spectrum of an operator is the part of its spectrum that remains invariant under compact perturbations).

Proof. Recall that $\mathbf{L}^t f$ is the difference between a multiplication operator and a compact operator

$$\mathbf{L}^t f(p) = g(p)f(p) - Kf$$

where $g(p) = (4\pi t)^{-\frac{k+2}{2}} \int_{\mathcal{M}} e^{-\frac{\|p-q\|^2}{4t}} d\mu_q$ and Kf is a convolution with a Gaussian. In [8] it is observed how, using basic perturbation theory, we can reach the result

$$SpecEss(\mathbf{L}^t) = \operatorname{rg} g$$

where $\operatorname{rg} g$ is the range of the function $g : \mathcal{M} \to \mathbb{R}$. Recall that $\operatorname{rg} g = \{g(p) \mid p \in M\} \subseteq \mathbb{R}$ i.e. the range is the set of all possible values that g can take. This fact is explained in more detail later in this section.

In order to estimate $\operatorname{rg} g$, observe first that

$$\lim_{t\to 0}(4\pi t)^{-\frac{k}{2}} \int_M e^{-\frac{\|p-q\|^2}{4t}} d\mu_q = 1.$$

Thus, for t small enough:

$$(4\pi t)^{-\frac{k}{2}} \int_M e^{-\frac{\|p-y\|^2}{4t}} d\mu_y > \frac{1}{2},$$

Recall that

$$g(p) = (4\pi t)^{-\frac{k+2}{2}} \int_{\mathcal{M}} e^{-\frac{\|p-q\|^2}{4t}} d\mu_q$$
$$= (4\pi t)^{-1}(4\pi t)^{-\frac{k}{2}} \int_{\mathcal{M}} e^{-\frac{\|p-q\|^2}{4t}} d\mu_q > (4\pi t)^{-1} \cdot \frac{1}{2}$$

Hence we have gotten that $g(p) > (4\pi t)^{-1} \cdot \frac{1}{2}$ and, since we are letting $t \to 0$, we can write that:

$$g(p) > \frac{1}{2}t^{-1}.$$

□

Lemma Let e^t be an eigenfunction of $\mathbf{L}^t$, satisfying

$$\mathbf{L}^t e^t = \lambda^t e^t, \quad \lambda^t < \frac{1}{2}t^{-1}.$$

Then $e^t \in C^\infty$.

From [14] we can make the following considerations:

If an operator is a compact perturbation of a multiplication operator, then any spectral values below the essential spectrum are isolated eigenvalues of finite multiplicity. In our case:

$$\mathbf{L}^t f(p) = g(p)f(p) - Kf,$$

Where K is a compact operator (convolution of a Gaussian). Compact operators only affect discrete spectrum (not essential spectrum), hence the only possible eigenvalues of $\mathbf{L}^t f(p)$ below $\frac{1}{2}t^{-1}$ are discrete eigenvalues.

From Fredhold Theory we know that for compact perturbations of self adjoint operators, the discrete eigenvalues must have finite dimensional eigenspaces. This means that for every eigenvalue $\lambda \in [0, \frac{1}{2}t^{-1}]$ the corresponding eigenfunctions form a finite dimensional subspace of the function space.

Let's now observe that $\hat{\mathbf{L}}_n^t$ and $\mathbf{L}^t$ are both bounded operators.

Proposition $\mathbf{L}^t$ is a bounded linear operator.

Proof. $\mathbf{L}^t f(p) = g(p)f(p) - Kf(p)$ is a linear operator as both its terms are linear in f.

We now want to prove it's bounded. Recall that an operator $T : C(\mathcal{M}) \to C(\mathcal{M})$ is bounded if there exist a constant $C > 0$ such that $||Tf||_\infty \leq C||f||_\infty$.

Observe that the operator $f \mapsto g(p)f(p)$ is bounded as $g(p) \in C^\infty(\mathcal{M})$ and $\mathcal{M}$ is compact. Thus the operator g is bounded.

We now show that the convolution operator Kf is bounded. Let $G(p, q)$ be the kernel of the operator K:

$$|Kf(p)| = \Big| \int_{\mathcal{M}} G(p,q)f(q)d\mu_q \Big| \leq \int_{\mathcal{M}} \big|G(p,q)\big| \cdot \big|f(q)\big| \, d\mu_q$$

By definition of infinity norm, for all $q \in \mathcal{M}$ we have that $|f(q)| \leq ||f||_\infty$. Thus

$$|K(f(p))| \leq ||f||_\infty \cdot \int_{\mathcal{M}} G(p,q)d\mu_q$$

Taking the supremum on both sides and using the fact that $||f||_\infty$ does not depend on p, we get that:

$$\sup_{p\in\mathcal{M}}|K(f(p))| \leq \sup_{p\in\mathcal{M}}\Big(||f||_\infty \cdot \int_{\mathcal{M}} G(p,q)d\mu_q\Big) \leq ||f||_\infty \cdot \sup_{p\in\mathcal{M}}\Big(\int_{\mathcal{M}} G(p,q)d\mu_q\Big)$$

Let's now define

$$C = \sup_{p\in\mathcal{M}} \int_{\mathcal{M}} G(p,q)d\mu_q$$

Since G is a continuous function on a compact set, then C is a (positive) finite number.

As $||Kf||_\infty = \sup_{p\in\mathcal{M}}|K(f(p))|$, this implies that

$$||Kf||_\infty \leq C||f||_\infty$$

□

Proposition $\hat{\mathbf{L}}_n^t$ is a bounded linear operator.

Proof. Recall that

$$\hat{\mathbf{L}}_n^t f(\mathbf{p}) = \frac{(4\pi t)^{-\frac{k+2}{2}}}{n}\Big(\sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}} f(\mathbf{p}) - \sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}} f(\mathbf{x}_i)\Big)$$

Observe that both terms of the operator only involve addition and multiplication, thus the operator is linear.

To show boundedness, let's apply the triangular inequality:

$$|\hat{\mathbf{L}}_n^t f(\mathbf{p})| \leq \frac{(4\pi t)^{-\frac{k+2}{2}}}{n}\Big(\sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}} |f(\mathbf{p})| + \sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}} |f(\mathbf{x}_i)|\Big)$$

Since the Gaussians $e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}}$ are both bounded and smooth, just as in the previous proof we can factor out $||f||_\infty$ from both sums, yielding:

$$|\hat{\mathbf{L}}_n^t f(\mathbf{p})| \leq C(p) \cdot ||f||_\infty$$

Where $C(p) = \frac{(4\pi t)^{-\frac{k+2}{2}}}{n} \sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}}$ Now, taking the supremum on both sides, we get:

$$||\hat{\mathbf{L}}_n^t f||_\infty \leq C \cdot ||f||_\infty$$

where $C = \sup_{p\in\mathcal{M}} C(p)$. Since $\sum_i e^{-\frac{||\mathbf{p}-\mathbf{x}_i||^2}{4t}}$ is a finite sum of smooth functions on a compact manifold, C is a finite positive constant.

□

Proposition $\hat{\mathbf{L}}_n^t$ converges uniformly to $\mathbf{L}^t$

Proof. We want to prove that for any continuous function $f \in C(\mathcal{M})$,

$$\|\hat{L}_n^t f - L^t f\|_\infty \to 0 \quad \text{as} \quad n \to \infty.$$

Let $C_t = (4\pi t)^{-\frac{k+2}{2}}$. Then:

$$\begin{aligned}
\left|\hat{\mathbf{L}}_n^t f(\mathbf{p}) - \mathbf{L}^t f(\mathbf{p})\right| &= C_t \left| \left(\frac{1}{n} \sum_i e^{-\frac{\|\mathbf{p}-\mathbf{x}_i\|^2}{4t}} f(\mathbf{p}) - \frac{1}{n} \sum_i e^{-\frac{\|\mathbf{p}-\mathbf{x}_i\|^2}{4t}} f(\mathbf{x}_i) \right) \right. \\
&\quad \left. - \left(\int_{\mathcal{M}} e^{-\frac{\|\mathbf{p}-\mathbf{q}\|^2}{4t}} f(\mathbf{p})\, d\mu_{\mathbf{q}} - \int_{\mathcal{M}} e^{-\frac{\|\mathbf{p}-\mathbf{q}\|^2}{4t}} f(\mathbf{q})\, d\mu_{\mathbf{q}} \right) \right| \\
&\leq C_t \left(\left| \frac{1}{n} \sum_i e^{-\frac{\|\mathbf{p}-\mathbf{x}_i\|^2}{4t}} - \int_{\mathcal{M}} e^{-\frac{\|\mathbf{p}-\mathbf{q}\|^2}{4t}}\, d\mu_{\mathbf{q}} \right| \cdot |f(\mathbf{p})| \right. \\
&\quad \left. + \left| \frac{1}{n} \sum_i e^{-\frac{\|\mathbf{p}-\mathbf{x}_i\|^2}{4t}} f(\mathbf{x}_i) - \int_{\mathcal{M}} e^{-\frac{\|\mathbf{p}-\mathbf{q}\|^2}{4t}} f(\mathbf{q})\, d\mu_{\mathbf{q}} \right| \right)
\end{aligned}$$

Let $K_t(\mathbf{p}, \mathbf{q}) = e^{-\frac{\|\mathbf{p}-\mathbf{q}\|^2}{4t}}$. Since $\mathcal{M}$ is compact and K_t is continuous, the functions:

$$\mathbf{q} \mapsto K_t(\mathbf{p}, \mathbf{q}) \quad \text{and} \quad \mathbf{q} \mapsto K_t(\mathbf{p}, \mathbf{q}) f(\mathbf{q})$$

are uniformly bounded and equicontinuous in $\mathbf{p}$. Then, by the Uniform Law of Large Numbers:

$$\sup_{\mathbf{p} \in \mathcal{M}} \left| \frac{1}{n} \sum_i K_t(\mathbf{p}, \mathbf{x}_i) - \int_{\mathcal{M}} K_t(\mathbf{p}, \mathbf{q})\, d\mu_{\mathbf{q}} \right| \to 0$$

$$\sup_{\mathbf{p} \in \mathcal{M}} \left| \frac{1}{n} \sum_i K_t(\mathbf{p}, \mathbf{x}_i) f(\mathbf{x}_i) - \int_{\mathcal{M}} K_t(\mathbf{p}, \mathbf{q}) f(\mathbf{q})\, d\mu_{\mathbf{q}} \right| \to 0$$

as $n \to \infty$, almost surely.

Since f is continuous on compact $\mathcal{M}$, it is bounded. Therefore, combining the two bounds:

$$\sup_{\mathbf{p} \in \mathcal{M}} \left| \hat{\mathbf{L}}_n^t f(\mathbf{p}) - \mathbf{L}^t f(\mathbf{p}) \right| \to 0 \quad \text{as } n \to \infty$$

□

Theorem The operator $\hat{\mathbf{L}}_n^t$ compactly converges to $\mathbf{L}^t$.

Proof. We recall that $\mathcal{M}$ is compact, and that for every $f \in C(\mathcal{M})$, we have already shown that $\hat{L}_n^t f$ converges uniformly to $L^t f$, i.e.:

$$\sup_{\mathbf{p}\in\mathcal{M}} \left| \hat{L}_n^t f(\mathbf{p}) - L^t f(\mathbf{p}) \right| \to 0 \quad \text{as} \quad n \to \infty.$$

By definition, compact convergence of a sequence of operators $\hat{L}_n^t$ to L^t means that for every $f \in C(\mathcal{M})$,

$$\|\hat{L}_n^t f - L^t f\|_\infty \to 0 \quad \text{as} \quad n \to \infty,$$

Thus, by the uniform convergence already established for each f, we conclude that $\hat{L}_n^t$ converges compactly to L^t. □

We now state Proposition 6. from [8] (proof can be found in [16]).

Proposition Let $(E, \|\cdot\|_E)$ be a complex Banach space, and let $(T_n)_{n\in\mathbb{N}}$ and T be bounded linear operators on E with $T_n \to T$ in the compact convergence sense. Let $\lambda \in \sigma(T)$ be an isolated eigenvalue with finite multiplicity m, and let $M \subset \mathbb{C}$ be an open neighborhood of λ such that

$$\sigma(T) \cap \overline{M} = \{\lambda\}.$$

Then:

- Convergence of eigenvalues: There exists an $N \in \mathbb{N}$ such that, for all $n > N$, the set $\sigma(T_n) \cap \overline{M}$ is an isolated part of $\sigma(T_n)$ consisting of at most m different eigenvalues, whose multiplicities sum up to m. Moreover, the sequence of sets $\sigma(T_n) \cap \overline{M}$ converges to the set $\{\lambda\}$ in the sense that every sequence $(\lambda_n)_{n\in\mathbb{N}}$ with $\lambda_n \in \sigma(T_n) \cap \overline{M}$ satisfies

$$\lim_{n\to\infty} \lambda_n = \lambda.$$

- Convergence of spectral projections: Let P be the spectral projection of T corresponding to λ, and for $n > N$, let P_n be the spectral projection of T_n corresponding to $\sigma(T_n) \cap \overline{M}$ (which is well-defined according to part 1). Then,

$$P_n \to P.$$

- Convergence of eigenvectors: If, additionally, λ is a simple eigenvalue, then there exists some $N \in \mathbb{N}$ such that, for all $n > N$, the set $\sigma(T_n) \cap \overline{M}$ consists of a single simple eigenvalue λ_n. The corresponding eigenfunctions f_n satisfy

$$f_n \to f \quad \text{up to multiplication by a unit complex number .}$$

  That is, there exists a sequence $(a_n)_{n\in\mathbb{N}}$ with $a_n \in \{-1, +1\}$ such that

$$a_n f_n \to f.$$

We have shown that the operators $\hat{L}_n^t$ converge compactly to L^t, and we established that the first eigenvalues are isolated with finite multiplicity. Furthermore, we verified that the operators $\hat{L}_n^t$ and L^t are bounded and linear. Thus applying the last proposition proves the spectral convergence of the empirical approximation to the functional approximation.

5 Laplacian Eigenmaps

The Laplacian Eigenmaps algorithm [1] addresses the problem of constructing a representations for data lying on a low-dimensional manifold embedded on a high dimensional space. In many areas of artificial intelligence one is often confronted with intrinsically low-dimensional data lying in a very high dimensional space. An example might be gray images of an object taken under fixed lighting conditions with a moving camera. Each such image, however, has intrinsic dimensionality equal to the degrees of freedom of the camera, but lies in the higher dimensional space that has as dimension the number of its pixels.

This algorithm approaches dimensionality reduction by building a graph that incorporates neighborhood information of the dataset. Then, it computes a low dimensional representation of the data that optimally preserves local neighborhood information.

5.1 The Algorithm

Given k points $\mathbf{x}_1, ..., \mathbf{x}_k \in \mathbb{R}^l$ we construct a weighted graph with k nodes, one for each point, and we set edges connecting neighboring points.

Step 1: Constructing the Adjacency Graph - We put an edge between nodes i and j if $\mathbf{x}_i$ and $\mathbf{x}_j$ are "close". We can define this "closeness" in two ways:

ε-neighbours (with $\varepsilon \in \mathbb{R}$) Nodes i and j are connected by an edge if $||\mathbf{x}_i - \mathbf{x}_j||^2 < \varepsilon$ where the norm is the usual euclidean norm in $\mathbb{R}^l$.

n nearest neighbors (with $n \in \mathbb{N}$) Nodes i and j are connected by an edge if i is among n nearest neighbors of j or vice-versa.

Step 2: Choosing the Weights - We have two ways to define the weighs:

Heat Kernel (with parameter $t \in \mathbb{R}$). If nodes i and j are connected we put $W_{ij} = e^{-\frac{||\mathbf{x}_i - \mathbf{x}_j||^2}{t}}$, otherwise $W_{ij} = 0$

No parameters ($t = \infty$) If nodes i and j are connected we put $W_{ij} = 1$ and $W_{ij} = 0$ otherwise.

Step 3: Eigenmaps - Assuming the graph G that we have constructed is connected (or proceed with this step for each connected component), we compute eigenvalues and eigenvectors for the generalized eigenvalue problem

$$L\mathbf{f} = \lambda D\mathbf{f}$$

Where D is diagonal weight matrix and its entries are column sums of W: $D_{ii} = \sum_j W_{ji}$. The matrix L is the Laplacian matrix and we have $L = D - W$. L is symmetric, positive semidefinite. It can be thought as an operator on functions defined on vertices of G.

Let $\mathbf{f}_0, ..., \mathbf{f}_{k-1}$ be the solution of the eigenvalue problem, ordered according to their eigenvalues:

$$L\mathbf{f}_0 = \lambda_0 D\mathbf{f}_0$$

$$L\mathbf{f}_1 = \lambda_1 D\mathbf{f}_1$$

$$\cdots$$

$$L\mathbf{f}_{k-1} = \lambda_{k-1} D\mathbf{f}_{k-1}$$

$$0 = \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_{k-1}$$

We leave out th eigenvector $\mathbf{f_0}$ cooresponding to the 0 eigenvalue, and use the next m eigevvectors for embedding in m-dimensional Euclidean space:

$$\mathbf{x}_i \mapsto (\mathbf{f}_1(i), ..., \mathbf{f}_m(i))$$

5.2 Justification

5.2.1 Optimal Embeddings

We first show that the algorithm preserves local information optimally. Given a dataset we have constructed the weighted graph as $G = (V, E)$ connecting nearby points to each other. Consider the simple case in which we want to map the graph to a line such that connected points will be mapped as close as possible. We let $\mathbf{y} = (y_1, y_2, ..., y_n)^T$ be that map.

To minimize the distance of connected points we can minimize the objective:

$$\sum_{ij}(y_i - y_j)^2 W_{ij}$$

with some appropriate constraints. It turns out that:

$$\frac{1}{2}\sum_{ij}(y_i - y_j)^2 W_{ij} = \mathbf{y}^T L\mathbf{y}$$

because in fact

$$\sum_{ij}(y_i - y_j)^2 W_{ij} = \sum_{ij}(y_i^2 - 2y_i y_j + y_j^2)W_{ij}$$

$$= \sum_i y_i^2 D_{ii} + \sum_j y_j^2 D_{jj} - 2\sum_{ij} W_{ij} 2y_i y_j = 2\mathbf{y}^T L\mathbf{y}$$

This calculation also shows that L is positive semidefinite. Therefore our minimization problem translates in:

$$\underset{\substack{\mathbf{y} \\ \mathbf{y}^T D\mathbf{y}=1}}{argmin}\ \mathbf{y}^T L\mathbf{y}$$

Were the constraint $\mathbf{y}^T D\mathbf{y} = 1$ removes the arbitrarily scaling factor in the embedding. Now, the vector $\mathbf{y}$ that minimizes the objective is given by the minimum eigenvalue solution to the generalized eigenvalue problem. Thus the solution is given by the smallest nonzero eigenvalue. In particular to delete the trivial solution with eigenvalue 0, we add the constraint $\mathbf{y}^T L\mathbf{1} = 0$ (because if the graph is connected $\mathbf{1}$ is the only eigenvector for $\lambda = 0$).

Now, if we were to consider the more general problem of embedding the graph to an m-dimensional Euclidean space, we would have an embedding given by a $k \times m$ matrix $\mathcal{Y} = [\mathbf{y}_1 \mathbf{y}_2 ... \mathbf{y}_m]$ where the ith row provides the embedding coordinates for the ith vertex. The objective to minimize will then be:

$$\sum_{ij} ||\mathbf{y}^{(i)} - \mathbf{y}^{(j)}||^2 W_{ij} = trace(\mathcal{Y}^T L \mathcal{Y})$$

Here $\mathbf{y}^{(i)} = [\mathbf{y}_1(i), ..., \mathbf{y}_m(i)]$ is the m-dimensional representationof the ith vertex. Thus we are reduced to find

$$\underset{\mathcal{Y}^T D \mathcal{Y} = I}{trace(\mathcal{Y}^T L \mathcal{Y})}$$

5.2.2 The Laplace Beltrami Operator

We consider $\mathcal{M}$ to be a smooth, compact, m -dimensional Riemannian Manifold. If the manifold is embedded in $\mathbb{R}^l$, the Riemann structure (metric tensor) on the manifold is induced by the standard Riemann structure on $\mathbb{R}^l$.

We want to provide a justification for why eigenfunctions of the Laplace Beltrami operators have desired properties for embeddings. To do so, as we have done for the graph Laplacian, we consider a map from the manifold to the real line, such that points close together on the manifold are mapped close together on the line. Let f be that map, and assume $f : \mathcal{M} \to \mathbb{R}$ is twice differentiable.

Let us consider two neighboring points $\mathbf{x}, \mathbf{z} \in \mathcal{M}$. They are mapped to $f(\mathbf{x})$ and $f(\mathbf{z})$.

We want to show that:

$$|f(\mathbf{z}) - f(\mathbf{x})| \leq dist_{\mathcal{M}}(\mathbf{x}, \mathbf{z}) ||\nabla f(\mathbf{x})|| + o(dist_{\mathcal{M}}(\mathbf{x}, \mathbf{z}))$$

Where $\nabla f(\mathbf{x})$ is the gradient. In particular the gradient is a vector of the tangent space $T\mathcal{M}_{\mathbf{x}}$ such that given another vector $\mathbf{v} \in T\mathcal{M}_{\mathbf{x}}$, we get $df(\mathbf{v}) = \langle \nabla f(x), \mathbf{v} \rangle_{\mathcal{M}}$. Now, we define $l = dist_{\mathcal{M}}(\mathbf{x}, \mathbf{z})$ and we let $c(t)$ be the geodesic curve parametrized by length connecting $\mathbf{x} = c(0)$ and $\mathbf{z} = c(l)$. Then observe:

$$f(\mathbf{z}) = f(\mathbf{x}) + \int_0^l df(c'(t))dt = f(\mathbf{x}) + \int_0^l df \langle \nabla f(c(t)), c'(t) \rangle dt$$

Now, using Cauchy-Schwartz inequality and the fact that $||c'(t)|| = 1$ (as it's a geodesic parametrized by length), we get:

$$\langle \nabla f(c(t)), c'(t) \rangle \leq ||\nabla f(c(t))||\ ||c'(t)|| = ||\nabla f(c(t))||$$

Then, by Taylor approximation:

$$||\nabla f(c(t))|| = ||\nabla f(\mathbf{x})|| + O(t)$$

Thus we are left with the inequality: $f(\mathbf{z}) \leq f(\mathbf{x}) + \int_0^l ||\nabla(f(\mathbf{x})|| + O(t)dt$ and by integrating, we get:

$$|f(\mathbf{z}) - f(\mathbf{x})| \leq l||\nabla(f(\mathbf{x})|| + o(l)$$

If $\mathcal{M}$ is isometrically embdedded in $\mathbb{R}^l$ then $dist_{\mathcal{M}}(\mathbf{x}, \mathbf{z}) = ||\mathbf{x} - \mathbf{z}||_{\mathbb{R}^l} + o(||\mathbf{x} - \mathbf{z}||_{\mathbb{R}^l})$, thus:

$$|f(\mathbf{z}) - f(\mathbf{x})| \leq ||\nabla(f(\mathbf{x})||\,||\mathbf{x} - \mathbf{z}|| + o(||\mathbf{x} - \mathbf{z}||)$$

Thus the gradient provides us with an estimate of how far apart f maps nearby points. Thus, looking for a map that best preserves locality, we will try to minimize $||\nabla(f(\mathbf{x})||$, and thus also $||\nabla(f(\mathbf{x})||^2$.

Our minimization problem will then be:

$$\underset{||f||_{L^2(\mathcal{M})=1}}{argmin} \int_{\mathcal{M}} ||\nabla(f(\mathbf{x})||^2$$

Although, the minimization problem also requires some constraints. In order to prevent arbitrary scaling we will need $||f||_{L^2(\mathcal{M})=1}$. Additionally we want the function to have 0 mean, thus we want it to integrate to 0.

Notice that minimizing the square of the gradient is equivalent to the objective that we have already used on the graph.

Using Stokes Theorem, we know that:

$$\int_{\mathcal{M}} ||\nabla(f(\mathbf{x})||^2 = \int_{\mathcal{M}} \Delta(f)f$$

Where Δ is the Laplace Beltrami operator. We can immediately notice that Δ is positive semidefinite, and that the solution of the minimization problem must be an eigenfunction of Δ.

We know the spectrum of the Laplace Beltrami operator on a compact manifold to be discrete, so we can write the eigenvalues in increasing order as:

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots$$

Let f_i be the eigenfunction corresponding to the eigenvalue λ_i. It it easy to see that f_0 is the constant function that maps the entire manifold to a single point. To avoid this solution we require the embedding map f to be orthogonal to f_0. It immediately follows that f_1 is the optimal embedding.

In particular, to find f_1 eigenfunction with eigenvalue λ_1 it is enough to restrict the minimization problem to the space orthogonal to f_0. In general, if we are looking for f_i eigenfunction with eigenvalue λ_i we can look for the space orthogonal to the functions we have already found.

Just as before, if we want to generalize to the embedding to an m-dimensional Euclidean space, we will have that the optimal embedding is:

$$\mathbf{x} \mapsto (f_1(\mathbf{x}), f_2(\mathbf{x}), ..., f_m(\mathbf{x}))$$

This last statement holds as eigenfunctions relative to different eigenvalues, as already mentioned above, are orthogonal to each other.

5.2.3 Heat Kernels and the Choice of Weight Matrix

The Laplace Beltrami operator is closely related o the heat flow. Let $f : \mathcal{M} \to \mathbb{R}$ be the initial heat distribution and let $u(x,t)$ be heat distribution at time t. We have that $u(x,0) = f(x)$. The heat equation is:

$$(\frac{\partial}{\partial t} + \Delta)u = 0$$

and it has solution:

$$u(x,t) = \int_{\mathcal{M}} H_t(x,y)f(y)$$

where H_t is the heat kernel. We then have:

$$\Delta f(x) = -\frac{\partial}{\partial t}u(x,0) = -\Big(\frac{\partial}{\partial t}\Big[\int_{\mathcal{M}} H_t(x,y)f(y)\Big]\Big)_{t=0}$$

Using the exponential map centered in $\mathbf{x}$, we can approximate H_t by a Gaussian:

$$H_t(x,y) = (4\pi t)^{-m/2} e^{-\frac{||x-y||^2}{4t}}(\phi(x,y) + O(t))$$

where $\phi(x,y)$ is a smooth function with $\phi(x,x) = 1$. Thus, when x and y are close enough:

$$H_t(x,y) \sim (4\pi t)^{-m/2} e^{-\frac{||x-y||^2}{4t}}$$

Notice that as t tends to 0 we have that the heat kernel tends to Dirac δ-function (i.e. $\lim_{t\to 0} \int_{\mathcal{M}} H_t(x,y)f(y) = f(x)$. Thus for t small enough and by definition of derivative we get:

$$\Delta f(x) \sim \frac{1}{t}\big[f(x) - (4\pi t)^{-m/2} \int_{\mathcal{M}} e^{-\frac{||x-y||^2}{4t}} f(y)dy\big]$$

In the discrete case (when $\mathbf{x}_1, ..., \mathbf{x}_k$ are datapoints on $\mathcal{M}$, this last expression can be approximated by:

$$\Delta f(\mathbf{x}_i) \sim \frac{1}{t}\big[f(\mathbf{x}_i) - \frac{1}{k}(4\pi t)^{-m/2} \sum_{\mathbf{x}_j : 0<||\mathbf{x}_j - \mathbf{x}_i||<\varepsilon} e^{-\frac{||\mathbf{x}_i-\mathbf{x}_j||^2}{4t}} f(\mathbf{x}_j)\big]$$

As the dimensionality of $\mathcal{M}$ might be unknown, we let $\alpha = \frac{1}{k}(4\pi t)^{-m/2}$.
Now, by observing that the Laplacian of the constant function is zero, it immediately follows that $\frac{1}{\alpha} \sim \sum_{\mathbf{x}_j : 0<||\mathbf{x}_j - \mathbf{x}_i||<\varepsilon} e^{-\frac{||\mathbf{x}_i-\mathbf{x}_j||^2}{4t}}$. This leads do some different approximations for the manifold Laplacian. Discarding the term α, which does not depend on the points, the most natural way to define weights (also assuring that the graph Laplacian is positive semidefinite) is the following

$$W_{ij} = \begin{cases} e^{-\frac{||\mathbf{x}_i-\mathbf{x}_j||^2}{4t}} \text{ if } ||\mathbf{x}_j - \mathbf{x}_i|| < \varepsilon \\ 0 \text{ otherwise} \end{cases}$$

5.3 Results

We show the effectiveness of the Laplacian Eigenmaps algorithm by sampling data points from 3D manifolds and embedding them into $\mathbb{R}^1$ and $\mathbb{R}^2$ using the algorithm. To illustrate the process, the original points are colored, allowing us to visually capture how the algorithm "unfolds" the complex shapes while preserving local geometric properties: the embeddings ensure that points close to one another on the manifold remain close in the lower-dimensional representations.

The following image represents and "S"-shaped manifold, and the relative result of Laplacian Eigenmaps algorithm using n-nearest neighbors for the closeness, and the parameter free weights. The last picture is the result of Principal Component Analysis on the manifold, to compare the difference between linear methods to the ones that preserve linearity only locally.

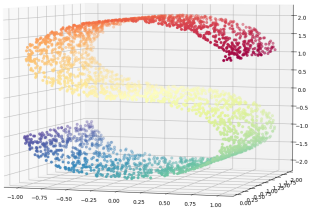


Figure 1: "S"-shaped manifold

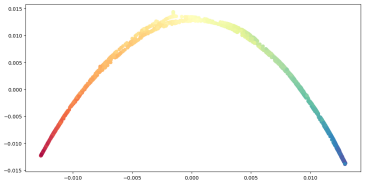


Figure 2: Laplacian Eigenmaps

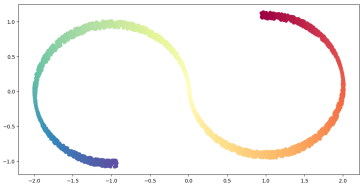


Figure 3: PCA

The same thing is done on a "sliced sphere".

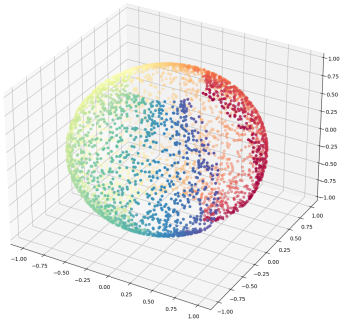


Figure 4: "sliced sphere"

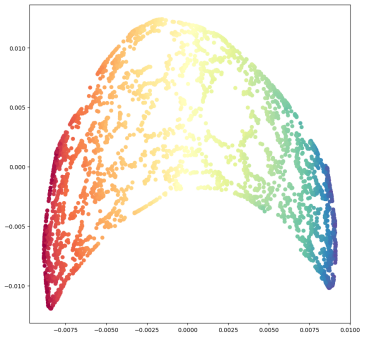


Figure 5: Laplacian Eigenmaps

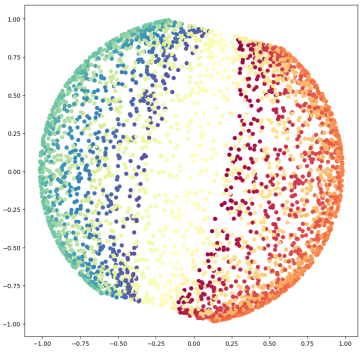


Figure 6: PCA

In particular this last example shows how, while PCA only projects the shape on the principal components, the Laplacian eigenmaps algorithm attempt to unfold the shape in such a way that "closeness" is only preserved locally.

Notice that in the justification of the algorithm we assumed (implicitly) that the Riemannian Manifold was compact and without boundary, while here the manifolds we have just showed do have boundary.
However, the relations we have used will still hold as long as it also holds the Neumann Condition (which is a type of boundary condition).

The Neumann condition specifies that the normal derivative of a function u on ∂M is zero:

$$\frac{\partial u}{\partial \nu} = \langle \nabla u, \nu \rangle_g = 0 \quad \text{on } \partial M,$$

where ν is the outward unit normal vector field on ∂M (with respect to g) and $\langle \nabla u, \nu \rangle_g$ is the inner product of ∇u and ν, representing the normal derivative of u at the boundary.

6 Coil-100 Dataset

The Columbia Object Image Library (COIL) is a dataset of color images of 100 different objects, each captured under various viewing angles. The objects were placed on a motorized turntable against a black background, and the turntable was rotated by fixed degrees to vary the object's pose relative to a stationary color camera. Images of each object were taken at pose intervals of 5 degrees, resulting in 72 images per object (360 degrees / 5 degrees = 72 images).



For each object, we apply the Laplacian Eigenmaps algorithm, to observe how well the object is embedded into $\mathbb{R}^3$ and $\mathbb{R}^2$. The success of this embedding varies depending on the geometry and symmetry of the object.

For instance, consider a Fortune cat. These toys typically have a distinct, asymmetrical design with clear differences between the front and back. This makes the embedding much more successful because the object's unique features lead to a well-separated and clear representation in the embedding space. Each pose taken from different angles will be distinctly identifiable, as the toy's features are easily distinguished.

On the other hand, a vase, which often has a symmetrical and simple shape, presents a more challenging embedding. Since the angles of the vase appear similar and its design is rotationally symmetric, the embeddings from different views can become more ambiguous, with images overlapping or becoming less distinct in the feature space.

In general, we would want the algorithm to recognize rotation by embedding the images into a structure that resembles a loop. This means that as an object rotates, the corresponding embeddings should form a closed curve, with the initial and final images coming together to indicate a complete rotation. Such a closed-loop structure not only reflects the cyclic nature of the rotation but also demonstrates that the algorithm effectively captures the underlying geometric transformation of the object's rotation.

6.1 Parameters and Choices for the Algorithm

As already mentioned in Section 5.1, the Laplacian Eigenmaps algorithm permits several choices for defining closeness between data points, selecting weights, and setting the associated parameters. In our experiments on this dataset, we tried various possibilities, and have observed that the best value for parameters are:

- $k = 10$ when choosing KNN as closeness criterion
- $\varepsilon = \mu$ where μ is the mean of the pairwise distances
- $t = 2\mu^2$ when we set the gaussian weights

Although alternative settings perform well on some images, we empirically find that the optimal configuration for this dataset is to use binary (0 or 1) weights combined with the k-nearest neighbors criterion for defining closeness. This is the setting we will use for the embeddings, as it provides the most consistent and effective results

6.2 Fortune Cat



As already mentioned, the object's asymmetry allows the algorithm to recognize the degrees of rotation. The following image shows the embedding of the object in $\mathbb{R}^2$.In particular, the points are colored according to the degrees of rotation as showed in the bar on the right; therefore, the more ordered they appear, the better the algorithm has performed.

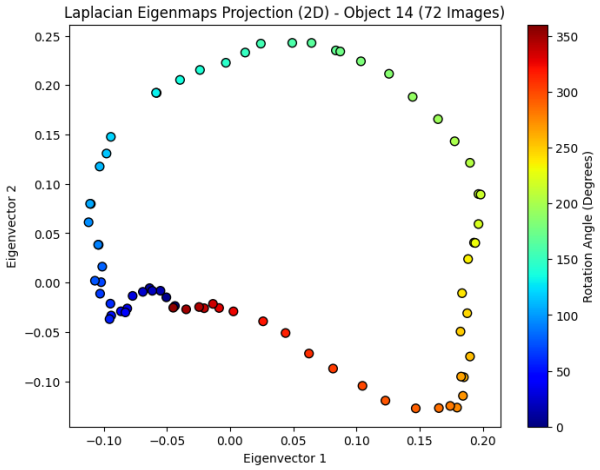


Figure 7: Object 14 (Fortune Cat) - embedded in $\mathbb{R}^2$

6.3 Vase



Object 25, the vase, is rotationally symmetrical, and as a consequence, its embedding does not exhibit a clear structure. Notice on the graph that there is no discernible pattern or closed-loop formation—even after experimenting with different parameter settings.

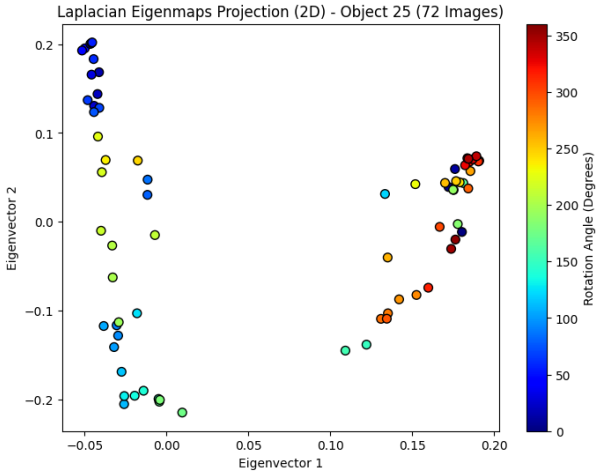


Figure 8: Object 25 (Vase) - embedded in $\mathbb{R}^2$

6.4 Car Toy



The embedding of the toy car images forms a clear loop, effectively capturing the continuous rotation of the object. However, it is noticeable that images corresponding to rotations 180° apart are embedded very close to each other. Observe, for example, second and last picture of the car toy. This is likely because the front and back of the car share similar visual features, causing the algorithm to confuse these views as being nearly identical. As a result, the Laplacian Eigenmaps method embeds these images in close proximity, reflecting the underlying symmetry in the car's design.

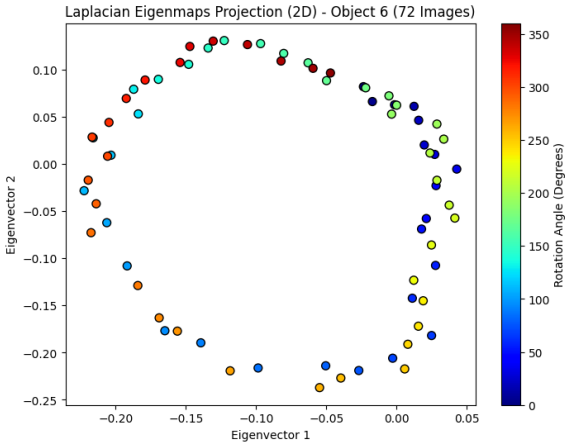


Figure 9: Object 6 (Car Toy) - embedded in $\mathbb{R}^2$

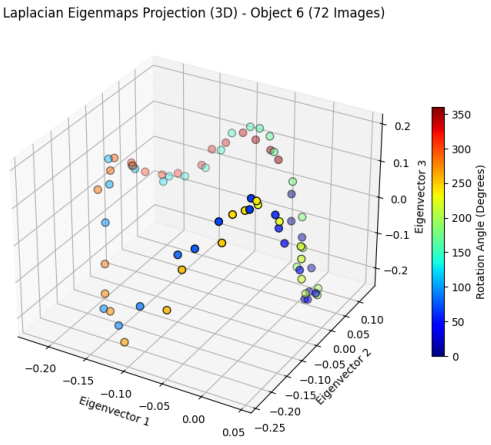
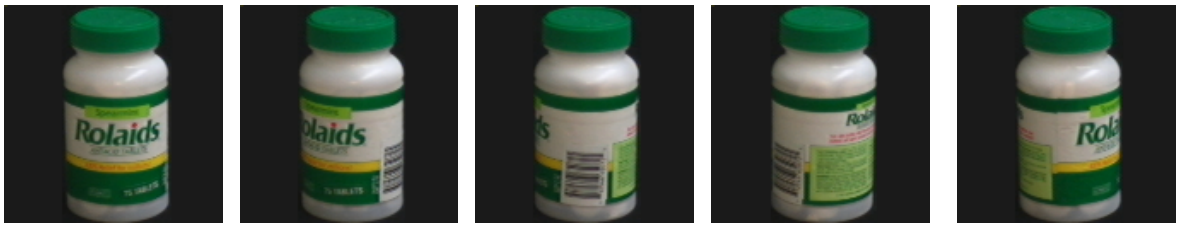


Figure 10: Object 6 (Car Toy) - embedded in $\mathbb{R}^3$

A for the embedding in $\mathbb{R}^2$ (on the left), it almost appears as if the loop has made an additional turn on itself, overlapping points that are 180° apart. This suggests that while the algorithm successfully captures the rotational structure, it does not fully distinguish between orientations that look visually similar, effectively "folding" the loop at the halfway mark of the rotation. On the right the picture of the embedding in $\mathbb{R}^3$, that, however, does not show better results.

A Appendix

The following appendix shows the Jupyter Notebook implementing the Laplacian Eigenmaps algorithm. At the beginning of the notebook, the object_label parameter can be modified to select the desired object. In this example, we selected object 5. The corresponding image is shown below.



Due to the symmetry of the object, and the presence of distinguishing labels on its sides, all the parameter choices explained in Section 5 perform reasonably well. However, the first method, which uses an unweighted matrix with k nearest neighbors, appears to yield the best result.

Since that method consistently produced better results across all images, it is also the one we chose to use in Section 6 to embed the various objects.

In [94]:
```
object_label = 5
```

K-nearest neighbors

In [95]:
```
import os
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
from sklearn.neighbors import kneighbors_graph
from scipy.sparse.linalg import eigsh
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.colors as mcolors



coil_100_dir = './coil-100/coil-100'


images = []
angles = []  #  rotation angles

#object_label = 10  # object number in the coil dataset
for degree in range(0, 360, 5):  #  0, 5, 10, ..., 355 degrees
    image_filename = f'obj{object_label}__{degree}.png'
    image_path = os.path.join(coil_100_dir, image_filename)

    if os.path.exists(image_path):
        img = Image.open(image_path).resize((128, 128))  # resolution (128x128)
        img_array = np.array(img).flatten()
        images.append(img_array)
        angles.append(degree)  # store the rotation angle as label


X_subset = np.array(images)
angles = np.array(angles)

print(images[0])


k = 10 #   neighbors
W = kneighbors_graph(X_subset, k, mode='connectivity', include_self=True).toarray()


D = np.diag(W.sum(axis=1))
L = D - W  # unnormalized Laplacian


n_components = 3  #
eigvals, eigvecs = eigsh(L, k=n_components+1, which='SM')  #  smallest eigenvalues
embedding = eigvecs[:, 1:]  # skiping  the first eigenvector (it's const func)


fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

# Normalize colors to span full range (0 to 360 degrees)
norm = mcolors.Normalize(vmin=0, vmax=360)
cmap = plt.get_cmap('jet')  # 'jet' provides full bright spectrum

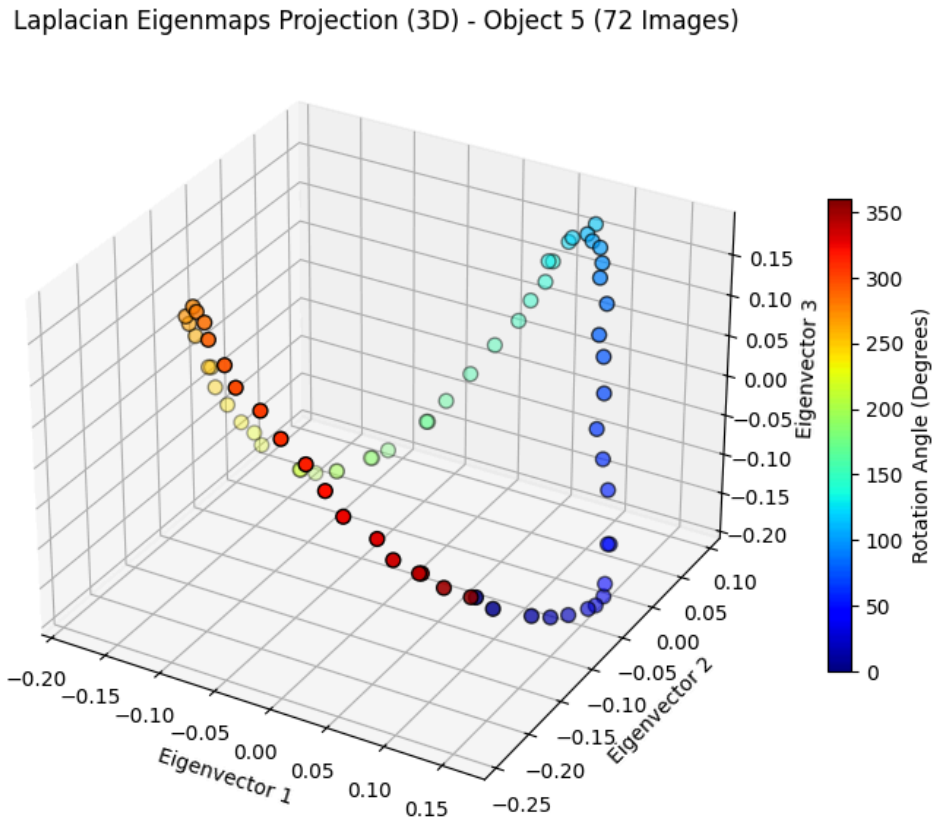
sc = ax.scatter(embedding[:, 0], embedding[:, 1], embedding[:, 2],
                c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)


cbar = plt.colorbar(sc, ax=ax, shrink=0.6)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (3D) - Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')
ax.set_zlabel('Eigenvector 3')

plt.show()
```

```
[53 51 32 ... 26 26 26]
```



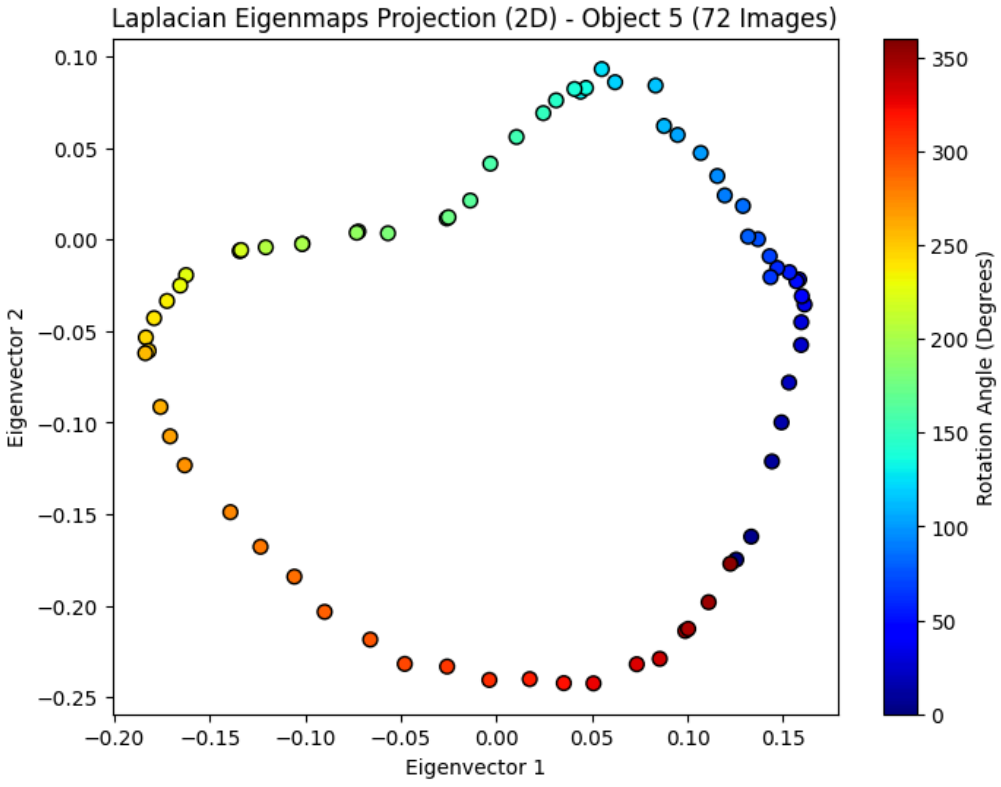
In [96]:
```
# 2D Projection
fig, ax = plt.subplots(figsize=(8, 6))

sc = ax.scatter(embedding[:, 0], embedding[:, 1], c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)

cbar = plt.colorbar(sc, ax=ax)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (2D) - Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')
```

Out[96]:
```
Text(0, 0.5, 'Eigenvector 2')
```



Weighted Matrix

In [97]:
```
from scipy.spatial.distance import cdist


# Euclidean distance matrix
pairwise_distances = cdist(X_subset, X_subset, metric='euclidean')

# affinity matrix using a heat kernel
sigma = np.mean(pairwise_distances)  # we set sigma as the mean distance
W = np.exp(-pairwise_distances**2 / (2 * sigma**2))

# laplacian
D = np.diag(W.sum(axis=1))
L = D - W  # unnormalized Laplacian

#  eigenvectors
n_components = 3  # dimensions for embedding
eigvals, eigvecs = eigsh(L, k=n_components+1, which='SM')  # compute smallest eigenvalues and eigenvectors
embedding = eigvecs[:, 1:]  # skip the first eigenvector (it's constant)


fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

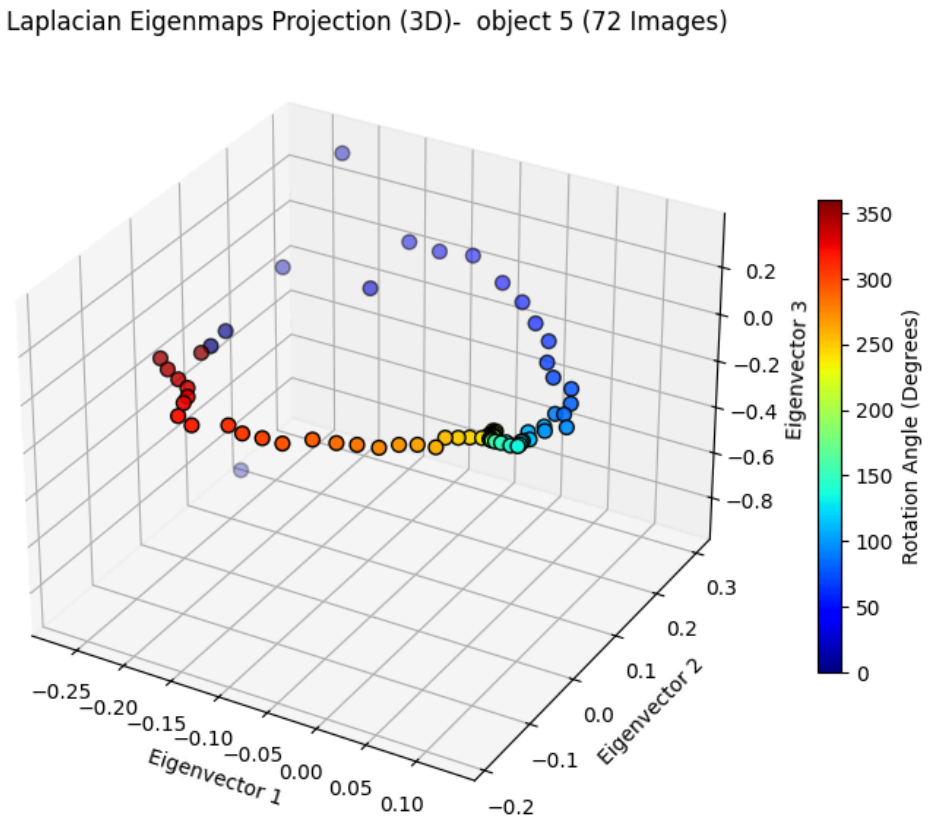

norm = mcolors.Normalize(vmin=0, vmax=360)
cmap = plt.get_cmap('jet')

sc = ax.scatter(embedding[:, 0], embedding[:, 1], embedding[:, 2],
                c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)

cbar = plt.colorbar(sc, ax=ax, shrink=0.6)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (3D)-  object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')
ax.set_zlabel('Eigenvector 3')

plt.show()
```



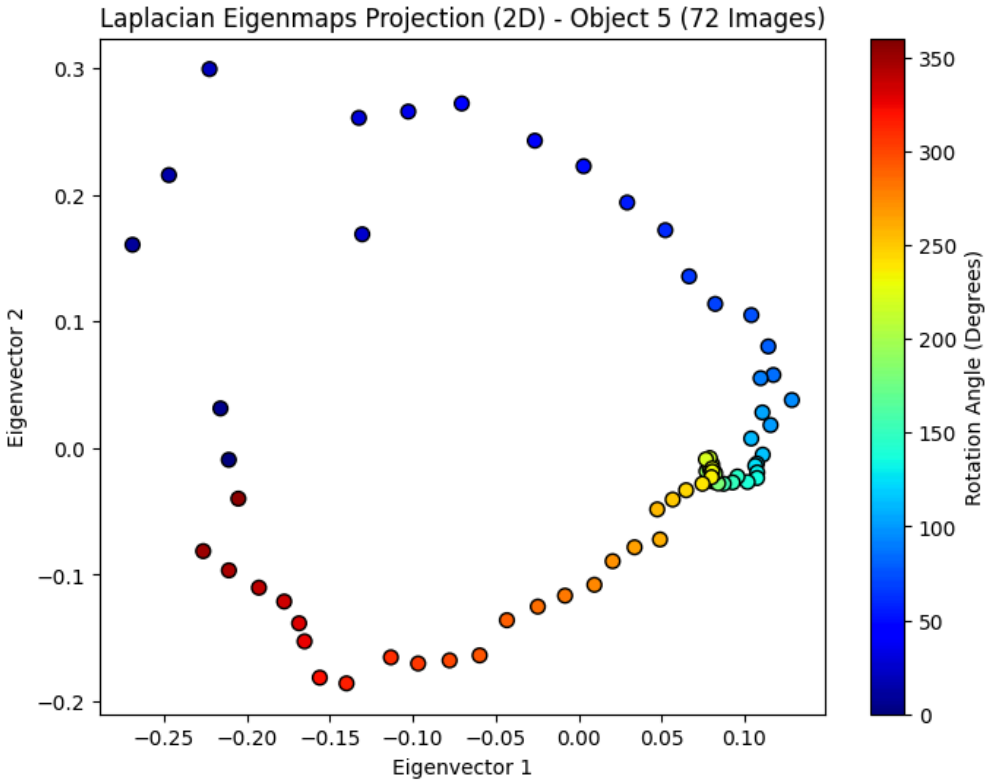
In [98]:
```
# 2D Projection
fig, ax = plt.subplots(figsize=(8, 6))

sc = ax.scatter(embedding[:, 0], embedding[:, 1], c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)
```

```
cbar = plt.colorbar(sc, ax=ax)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (2D) - Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')

plt.show()
```



Weighted Matrix with ε treshold

In [99]:

```
pairwise_distances = cdist(X_subset, X_subset, metric='euclidean')



epsilon =  np.mean(pairwise_distances)  # we set epsilon as the mean distance
print("epsiolon", epsilon)
W = np.exp(-pairwise_distances**2 / (2 * epsilon**2))
W[pairwise_distances > epsilon] = 0


D = np.diag(W.sum(axis=1))
L = D - W

n_components = 3
eigvals, eigvecs = eigsh(L, k=n_components+1, which='SM')
embedding = eigvecs[:, 1:]


fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

norm = mcolors.Normalize(vmin=0, vmax=360)
cmap = plt.get_cmap('jet')

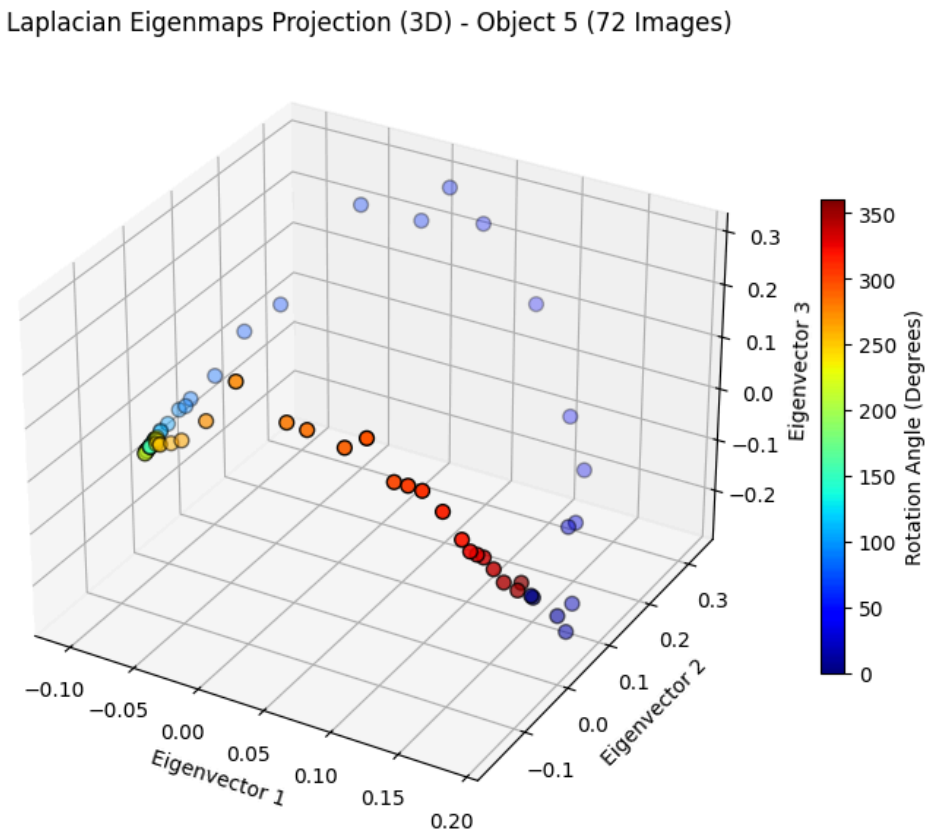
sc = ax.scatter(embedding[:, 0], embedding[:, 1], embedding[:, 2],
                c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)

cbar = plt.colorbar(sc, ax=ax, shrink=0.6)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (3D) - Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')
ax.set_zlabel('Eigenvector 3')
```

```
plt.show()
```

```
epsiolon 4375.917053570752
```



In [100…

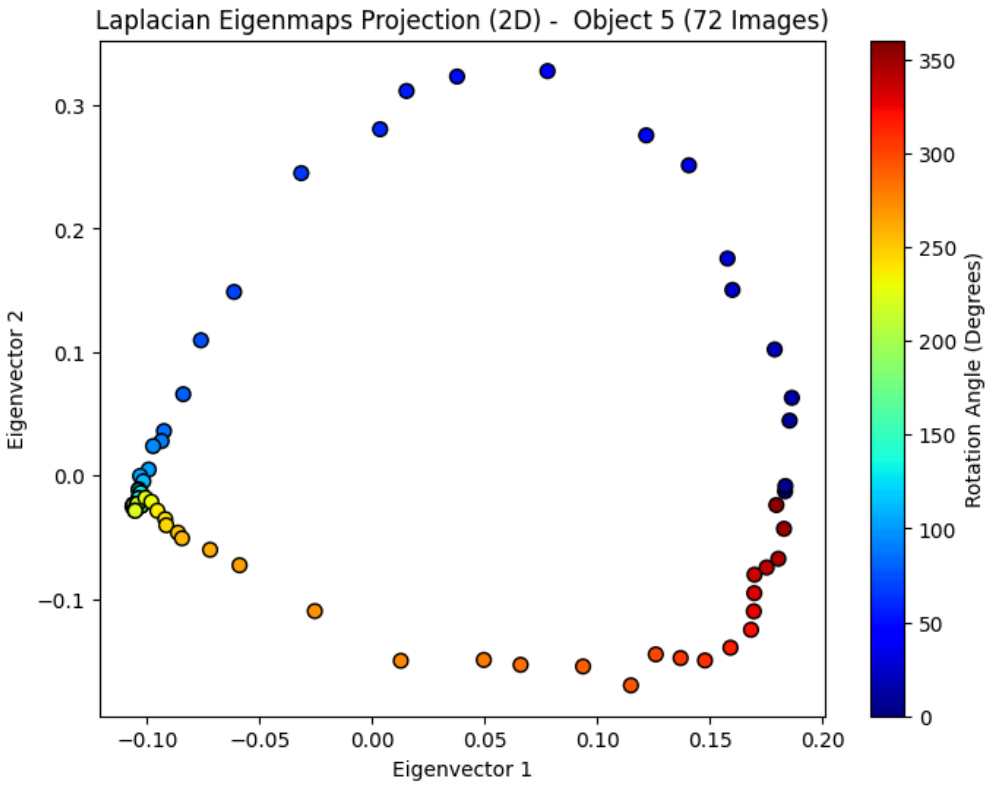
```
# 2D Projection
fig, ax = plt.subplots(figsize=(8, 6))

sc = ax.scatter(embedding[:, 0], embedding[:, 1], c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)

cbar = plt.colorbar(sc, ax=ax)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (2D) -  Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')

plt.show()
```



Non-weighted Matrix with ε treshold

In [101…

```
pairwise_distances = cdist(X_subset, X_subset, metric='euclidean')

epsilon = np.mean(pairwise_distances)
W = (pairwise_distances < epsilon).astype(float)

D = np.diag(W.sum(axis=1))
L = D - W

n_components = 3
eigvals, eigvecs = eigsh(L, k=n_components+1, which='SM')
embedding = eigvecs[:, 1:]

fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

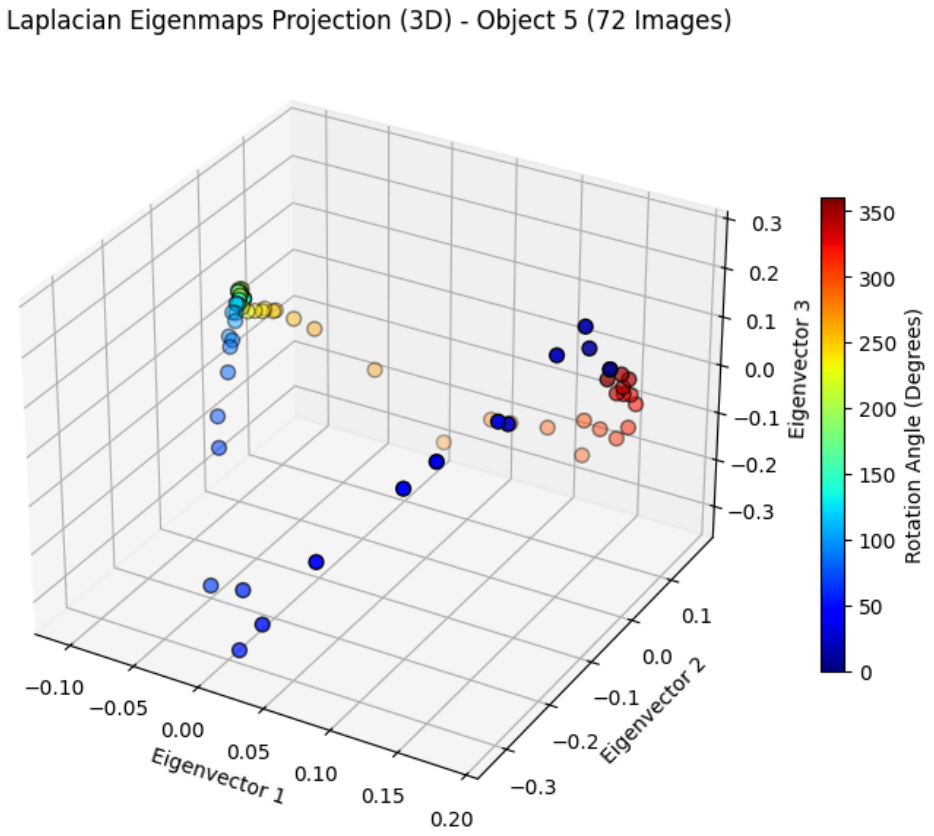
norm = mcolors.Normalize(vmin=0, vmax=360)
cmap = plt.get_cmap('jet')

sc = ax.scatter(embedding[:, 0], embedding[:, 1], embedding[:, 2],
                c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)

cbar = plt.colorbar(sc, ax=ax, shrink=0.6)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (3D) - Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')
ax.set_zlabel('Eigenvector 3')

plt.show()
```



In [102…

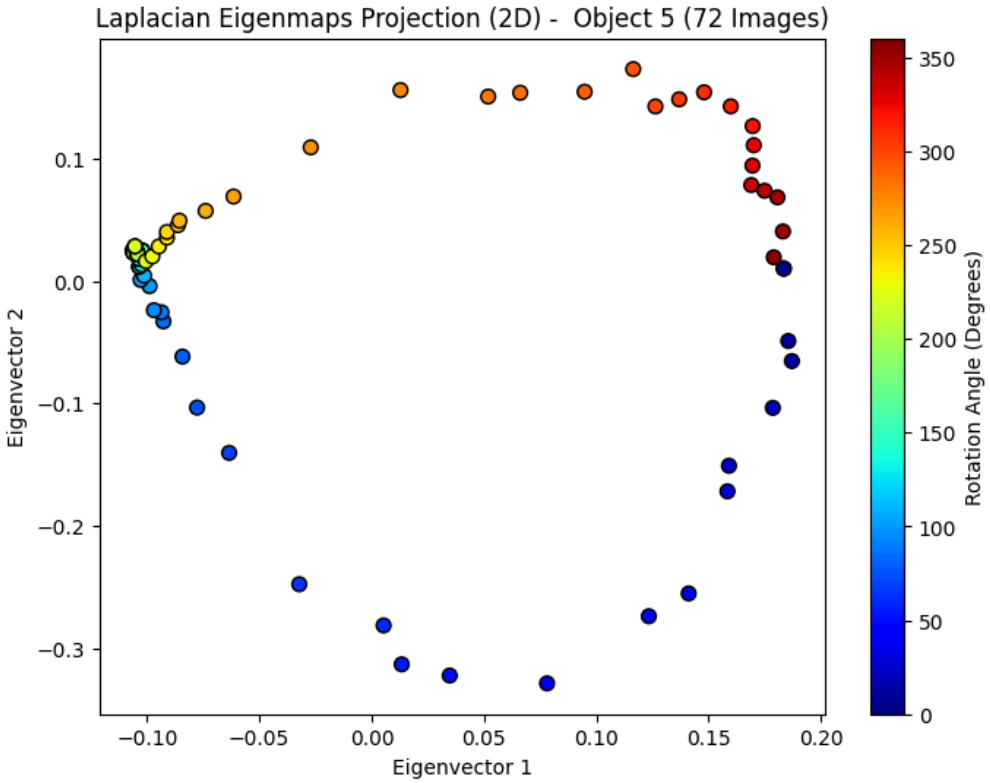
```
# 2D Projection
fig, ax = plt.subplots(figsize=(8, 6))

sc = ax.scatter(embedding[:, 0], embedding[:, 1], c=angles, cmap=cmap, norm=norm, edgecolor='k', s=50)

cbar = plt.colorbar(sc, ax=ax)
cbar.set_label('Rotation Angle (Degrees)')

ax.set_title('Laplacian Eigenmaps Projection (2D) -  Object ' + str(object_label) +' (72 Images)')
ax.set_xlabel('Eigenvector 1')
ax.set_ylabel('Eigenvector 2')

plt.show()
```



References

[1] M. Belkin, P. Niyogi. *Laplacian eigenmaps for dimensionality reduction and data Representation*. Neural Computations, 2002.

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